# 1. Overview

The MCU integrates multiple series of software-compatible Arm®-based 32-bit cores that share a common set of Renesas peripherals to facilitate design scalability and efficient platform-based product development.

The MCU in this series incorporates a high-performance Arm® Cortex®-M85 core running up to 1 GHz and Arm® Cortex®-M33 core running up to 250 MHz with the following features:

- Up to 1 MB MRAM
- 2 MB SRAM (256 KB of CM85 TCM RAM, 128 KB CM33 TCM RAM, 1664 KB of user SRAM)
- Octal Serial Peripheral Interface (OSPI)
- Layer 3 Ethernet Switch Module (ESWM), USBFS, SD/MMC Host Interface
- Analog peripherals
- Security and safety features

## 1.1 Function Outline

**Table 1.1 Arm core**

| Feature | Functional description |
|---|---|
| Arm® Cortex®-M85 core | • Maximum operating frequency: up to 1 GHz<br>• Arm® Cortex®-M85 core<br>– Revision: (r1p1-00rel0)<br>– ARMv8.1-M architecture profile<br>– Armv8-M Security Extension<br>– Floating Point Unit (FPU) compliant with the ANSI/IEEE Std 754-2008 Scalar half, single, and double-precision floating-point operation<br>– M-profile Vector Extension (MVE) Integer, half-precision, and single-precision floating-point MVE (MVE-F)<br>• Arm® Memory Protection Unit (Arm MPU)<br>– Protected Memory System Architecture (PMSAv8)<br>– Secure MPU (MPU_S): 8 regions<br>– Non-secure MPU (MPU_NS): 8 regions<br>• SysTick timer<br>– Embeds two Systick timers: Secure instance (SysTick_S) and Non-secure instance (SysTick_NS)<br>– Driven by CPUCLK0 or MOCO divided by 8<br>• CoreSight™ ETM-M85 |
| Arm® Cortex®-M33 core | • Maximum operating frequency: up to 250 MHz<br>• Arm® Cortex®-M33 core<br>– Revision: (r0p4-00rel2)<br>– ARMv8-M architecture profile<br>– Armv8-M Security Extension<br>– Armv8-DSP Extension<br>– Floating Point Unit (FPU) compliant with the ANSI/IEEE Std 754-2008 single-precision floating-point operation<br>• Arm® Memory Protection Unit (Arm MPU)<br>– Protected Memory System Architecture (PMSAv8)<br>– Secure MPU (MPU_S): 8 regions<br>– Non-secure MPU (MPU_NS): 8 regions<br>• SysTick timer<br>– Embeds two Systick timers: Secure instance (SysTick_S) and Non-secure instance (SysTick_NS)<br>– Driven by CPUCLK1 or MOCO divided by 8<br>• CoreSight™ ETM-M33 |

**Table 1.2 Memory (1 of 2)**

| Feature | Functional description |
|---|---|
| Code MRAM | Maximum 1 MB of code MRAM.<br>See section 58, MRAM. |
| Flash memory | System in package (SiP) maximum 8 MB serial flash memory.<br>See section 44.4. SiP Product Configuration. |

**Table 1.2 Memory (2 of 2)**

| Feature | Functional description |
|---|---|
| Option-setting memory | The option-setting memory determines the state of the MCU after a reset.<br>See section 7, Option-Setting Memory. |
| SRAM | On-chip high-speed SRAM with Error Correction Code (ECC).<br>See section 57, SRAM. |
| OTP | On-chip OTP contains First Stage Bootloader (FSBL)<br>General purpose 96-byte OTP |

**Table 1.3 System**

| Feature | Functional description |
|---|---|
| Operating modes | Three operating modes:<br>● Single-chip mode<br>● JTAG boot mode<br>● SCI/USB boot mode<br>See section 4, Operating Modes. |
| Resets | This MCU provides the following 21 types of reset.<br>See section 6, Resets. |
| Programmable Voltage Detection (PVD) | The Programmable Voltage Detection (PVD) module monitors the voltage level input to the VCC pin. The detection level can be selected by register settings. The PVD module consists of five separate voltage level detectors (PVD0, PVD1, PVD2, PVD4, PVD5). These PVDs measure the voltage level input to the VCC pin. PVD registers allow your application to configure detection of VCC changes at various voltage thresholds.<br>See section 8, Programmable Voltage Detection (PVD). |
| Clocks | ● Main clock oscillator (MOSC)<br>● Sub-clock oscillator (SOSC)<br>● High-speed on-chip oscillator (HOCO)<br>● Middle-speed on-chip oscillator (MOCO)<br>● Low-speed on-chip oscillator (LOCO)<br>● PLL1/PLL2<br>● Clock out support<br>See section 9, Clock Generation Circuit. |
| Clock Frequency Accuracy Measurement Circuit (CAC) | The Clock Frequency Accuracy Measurement Circuit (CAC) counts pulses of the clock to be measured (measurement target clock) within the time generated by the clock selected as the measurement reference (measurement reference clock), and determines the accuracy depending on whether the number of pulses is within the allowable range. When measurement is complete or the number of pulses within the time generated by the measurement reference clock is not within the allowable range, an interrupt request is generated.<br>See section 10, Clock Frequency Accuracy Measurement Circuit (CAC). |
| Interrupt Controller Unit (ICU) | The Interrupt Controller Unit (ICU) controls which event signals are linked to the Nested Vector Interrupt Controller (NVIC), DMA Controller (DMAC) module and the Data Transfer Controller (DTC) modules. The ICU also controls non-maskable interrupts.<br>See section 14, Interrupt Controller Unit (ICU). |
| Low power modes | Power consumption can be reduced in multiple ways, including setting clock dividers, controlling EBCLK output, controlling SDCLK output, stopping modules, power gating control, selecting operating power control modes in normal operation, and transitioning to low power modes and processor low power modes.<br>See section 11, Low Power Mode. |
| Battery backup function | A battery backup function is provided for partial powering by a battery. The battery-powered area includes the RTC, SOSC, backup register, tamper detection and VBATT_R voltage drop detection and switch between VCC and VBATT.<br>See section 12, Battery Backup Function. |
| Register write protection | The register write protection function protects important registers from being overwritten due to software errors. The registers to be protected are set with the Protect Register (PRCR_S and PRCR_NS).<br>See section 13, Register Write Protection Function. |
| Memory Protection Unit (MPU) | All bus masters have Memory Protection Units (MPUs).<br>See section 16, Memory Protection Unit (MPU). |

**Table 1.4 Event link**

| Feature | Functional description |
|---|---|
| Event Link Controller (ELC) | The Event Link Controller (ELC) uses the event requests generated by various peripheral modules as source signals to connect them to different modules, allowing direct link between the modules without CPU intervention.<br>See section 19, Event Link Controller (ELC). |

**Table 1.5 Direct memory access**

| Feature | Functional description |
|---|---|
| Data Transfer Controller (DTC) | A Data Transfer Controller (DTC) module is provided for transferring data when activated by an interrupt request.<br>See section 18, Data Transfer Controller (DTC). |
| DMA Controller (DMAC) | The 8-channel direct memory access controller (DMAC) that can transfer data without intervention from the CPU. When a DMA transfer request is generated, the DMAC transfers data stored at the transfer source address to the transfer destination address.<br>See section 17, DMA Controller (DMAC). |

**Table 1.6 External bus interface**

| Feature | Functional description |
|---|---|
| External buses | ● CS area (ECBI): Connected to the external devices (external memory interface)<br>● SDRAM area (ECBI): Connected to the SDRAM (external memory interface)<br>● OSPI0 area (OSPI0BI): Connected to the OSPI0 (external device interface)<br>● OSPI1 area (OSPI1BI): Connected to the OSPI1 (external device interface) |

**Table 1.7 Timers (1 of 2)**

| Feature | Functional description |
|---|---|
| General PWM Timer (GPT) | The General PWM Timer (GPT) is a 32-bit timer with GPT32 × 14 channels. PWM waveforms can be generated by controlling the up-counter, down-counter, or the up- and down-counter. In addition, PWM waveforms can be generated for controlling brushless DC motors. The GPT can also be used as a general-purpose timer.<br>See section 22, General PWM Timer (GPT). |
| PWM Delay Generation Circuit (PDG) | The PWM Delay Generation circuit (PDG) has 4 channels delay circuits that can connect to the GPT. The PDG can control the rise and fall edge timing with which the PWM output for the GPT320 through the GPT323.<br>See section 23, PWM Delay Generation Circuit (PDG). |
| Port Output Enable for GPT (POEG) | The Port Output Enable (POEG) function can place the General PWM Timer (GPT) output pins in the output disable state<br>See section 21, Port Output Enable for GPT (POEG). |
| Low Power Asynchronous General Purpose Timer (AGT) | The Low Power Asynchronous General Purpose Timer (AGT) is a 16-bit timer that can be used for pulse output, external pulse width or period measurement, and counting external events. This timer consists of a reload register and a down counter. The reload register and the down counter are allocated to the same address, and can be accessed with the AGT register.<br>See section 24, Low Power Asynchronous General Purpose Timer (AGT). |
| Ultra-Low-Power Timer (ULPT) | The Ultra-Low-Power Timer (ULPT) is a 32-bit timer which can be used for outputting pulses or counting external events. This 32-bit timer consists of reload registers and a down-counter. The reload registers and the down-counter are allocated to the same address and can be accessed through the ULPTCNT register.<br>See section 25, Ultra-Low-Power Timer (ULPT) |
| Realtime Clock (RTC) | The realtime clock (RTC) has two counting modes, calendar count mode and binary count mode, that are used by switching register settings. For calendar count mode, the RTC has a 100-year calendar from 2000 to 2099 and automatically adjusts dates for leap years. For binary count mode, the RTC counts seconds and retains the information as a serial value. Binary count mode can be used for calendars other than the Gregorian (Western) calendar.<br>See section 26, Realtime Clock (RTC). |
| Watchdog Timer (WDT) | The Watchdog Timer (WDT) is a 14-bit down counter that can be used to reset the MCU when the counter underflows because the system has run out of control and is unable to refresh the WDT. In addition, the WDT can be used to generate a non-maskable interrupt or an underflow interrupt.<br>See section 27, Watchdog Timer (WDT). |

**Table 1.7 Timers (2 of 2)**

| Feature | Functional description |
|---|---|
| Independent Watchdog Timer (IWDT) | The Independent Watchdog Timer (IWDT) has a 14-bit down-counter, which resets the MCU by a reset output when the down-counter underflows. Alternatively, generation of an interrupt request when the counter underflows can be selected. This enables detection of a program runaway taking the refresh interval into account. The IWDT has two start modes: auto start mode, in which counting automatically starts after release from the reset state, and register start mode, in which counting is started by refreshing (writing to a specific register).<br>See section 28, Independent Watchdog Timer (IWDT). |

**Table 1.8 Communication interfaces (1 of 2)**

| Feature | Functional description |
|---|---|
| Serial Communications Interface (SCI) | The Serial Communications Interface (SCI) × 10 channels have asynchronous and synchronous serial interfaces:<br>● Asynchronous interfaces (UART and Asynchronous Communications Interface Adapter (ACIA))<br>● 8-bit clock synchronous interface<br>● Simple IIC (master-only)<br>● Simple SPI<br>● Smart card interface<br>● Manchester interface<br>● Simple LIN interface<br>The smart card interface complies with the ISO/IEC 7816-3 standard for electronic signals and transmission protocol. All channels have FIFO buffers to enable continuous and full-duplex communication, and the data transfer speed can be configured independently using an on-chip baud rate generator.<br>The maximum rate supported on this MCU. Refer to the electrical characteristics for the actual rate.<br>See section 38, Serial Communications Interface (SCI). |
| $I^2C$ Bus interface (IIC) | The $I^2C$ Bus interface (IIC) has 3 channels. The IIC module conforms with and provides a subset of the NXP $I^2C$ (Inter-Integrated Circuit) bus interface functions.<br>See section 39, $I^2C$ Bus Interface (IIC). |
| I3C Bus Interface (I3C) | The I3C Bus Interface (I3C) has 1 channel. The I3C module conform with and provide a subset of the NXP $I^2C$ (Inter-Integrated Circuit) bus interface functions and a subset of the MIPI I3C.<br>See section 40, I3C Bus Interface (I3C). |
| Serial Peripheral Interface (SPI) | The Serial Peripheral Interface (SPI) provides high-speed full-duplex synchronous serial communications with multiple processors and peripheral devices.<br>The maximum rate supported on this MCU. Refer to the electrical characteristics for the actual rate.<br>See section 43, Serial Peripheral Interface (SPI). |
| Control Area Network with Flexible Data-Rate Module (CANFD) | The CAN with Flexible Data-Rate (CANFD) module can handle classical CAN frames and CANFD frames complied with ISO 11898-1 standard.<br>The module supports 4 transmit buffers per channel and 16 receive buffers per channel.<br>See section 41, CAN with Flexible Data-rate (CANFD). |
| USB 2.0 Full-Speed module (USBFS) | The USB 2.0 Full-Speed module (USBFS) can operate as a host controller or device controller. The module supports full-speed and low-speed (host controller only) transfer as defined in Universal Serial Bus Specification 2.0. The module has an internal USB transceiver and supports all of the transfer types defined in Universal Serial Bus Specification 2.0. The USB has buffer memory for data transfer, providing a maximum of 10 pipes. Pipes 1 to 9 can be assigned any endpoint number based on the peripheral devices used for communication or based on your system.<br>See section 37, USB 2.0 Full-Speed Module (USBFS). |
| Octal Serial Peripheral Interface (OSPI) | The Octal Serial Peripheral Interface (OSPI) is a memory controller that supports Expanded Serial Peripheral Interface (xSPI) (JEDEC Standard JESD251, JESD251-1 and JESD252). The OSPI supports 1-bit, 2-bit, 4-bit and 8-bit protocols.<br>JESD251 specifies two interface profiles where profile 1.0 is Octal SPI and profile 2.0 is HyperBus™ (HyperRAM™ and HyperFlash™). OSPI supports QSPI protocol.<br>See section 44, Octal Serial Peripheral Interface (OSPI). |

**Table 1.8 Communication interfaces (2 of 2)**

| Feature | Functional description |
|---|---|
| SD/MMC Host Interface (SDHI) | The Secure Digital (SD) Card and Multi Media Card (MMC) Host Interface provides the functionality required to connect a variety of external memory cards to the MCU. The SDHI supports both 1- and 4-bit buses for connecting memory cards that support SD, SDHC, and SDXC formats. When developing host devices that are compliant with the SD Specifications, you must comply with the SD Host/Ancillary Product License Agreement (SD HALA). The MMC interface supports 1-bit, 4-bit, and 8-bit MMC buses that provide eMMC 4.51 (JEDEC Standard JESD 84-B451) device access. This interface also provides backward compatibility and supports high-speed SDR transfer modes.<br>See section 46, SD/MMC Host Interface (SDHI). |
| Layer 3 Ethernet Switch Module (ESWM) | The Layer 3 Ethernet Switch Module (ESWM) consists of two channels of Gigabit Ethernet controller, an Ethernet switch with high level routing capability, and multi-protocol interface support. The Gigabit Ethernet controller conforms to the definition of the Ethernet MAC (Media Access Control) layer in the IEEE 802.3 standard. This can transmit and receive Ethernet (IEEE 802.3) frames by connecting with an external physical-layer LSI chip (PHY-LSI) which complies with the standard. The Ethernet switch allows autonomous frame routing within a same network interface protocol, or between different network interfaces protocols or optimized gateway applications.<br>See section 29, Layer 3 Ethernet Switch Module (ESWM). |
| EtherCAT Slave Controller (ESC) | The EtherCAT slave controller (ESC) uses an EtherCAT Slave Controller IP Core (ESC IP Core) made by Beckhoff Automation GmbH, Germany. The ESC handles the EtherCAT communication as an interface between the EtherCAT fieldbus and the slave application.<br>See section 36, EtherCAT Slave Controller (ESC). |
| Delta-Sigma Modulator Interface (DSMIF) | DSMIF has three channels that are connectable with external delta-sigma modulators. DSMIF can be connected with up to three external delta-sigma modulators. Each DSMIF can filter and convert 1-bit digital data streams that were delta-sigma modulated at a high sampling rate into 16-bit digital data at a lower sampling rate.<br>See section 48, ΔΣ Interface (DSMIF). |

**Table 1.9 Analog**

| Feature | Functional description |
|---|---|
| 16-bit A/D Converter (ADC16H) | A 16-bit A/D Converter is provided. Up to 23 analog input channels are selectable. Temperature sensor output, and internal reference voltage and VBATT 1/6 voltage monitor are selectable for conversion.<br>See section 52, 16-bit A/D Converter (ADC16H). |
| 12-bit D/A Converter (DAC12) | A 12-bit D/A Converter (DAC12) is provided.<br>See section 53, 12-Bit D/A Converter (DAC12). |
| Temperature Sensor (TSN) | The on-chip Temperature Sensor (TSN) determines and monitors the die temperature for reliable operation of the device. The sensor outputs a voltage directly proportional to the die temperature, and the relationship between the die temperature and the output voltage is fairly linear. The output voltage is provided to the ADC16H for conversion and can be further used by the end application. The sensor outputs an abnormal temperature detection signal to the reset control circuit and can be used to prevent the malfunction due to abnormal temperature.<br>See section 54, Temperature Sensor (TSN). |
| High-Speed Analog Comparator (ACMPHS) | The High-Speed Analog Comparator (ACMPHS) can be used to compare an analog input voltage with a reference voltage and to provide a digital output based on the result of conversion. Both the analog input voltage and the reference voltage can be provided to the ACMPHS from internal sources (D/A converter output or internal reference voltage) and an external source. Such flexibility is useful in applications that require go/no-go comparisons to be performed between analog signals without necessarily requiring A/D conversion.<br>See section 55, High-Speed Analog Comparator (ACMPHS). |

**Table 1.10 Data processing (1 of 2)**

| Feature | Functional description |
|---|---|
| Cyclic Redundancy Check (CRC) calculator | The Cyclic Redundancy Check (CRC) calculator generates CRC codes to detect errors in the data. The bit order of CRC calculation results can be switched for LSB-first or MSB-first communication. Additionally, various CRC-generation polynomials are available. The snoop function allows monitoring reads from and writes to specific addresses. This function is useful in applications that require CRC code to be generated automatically in certain events, such as monitoring writes to the serial transmit buffer and reads from the serial receive buffer.<br>See section 47, Cyclic Redundancy Check (CRC). |

**Table 1.10 Data processing (2 of 2)**

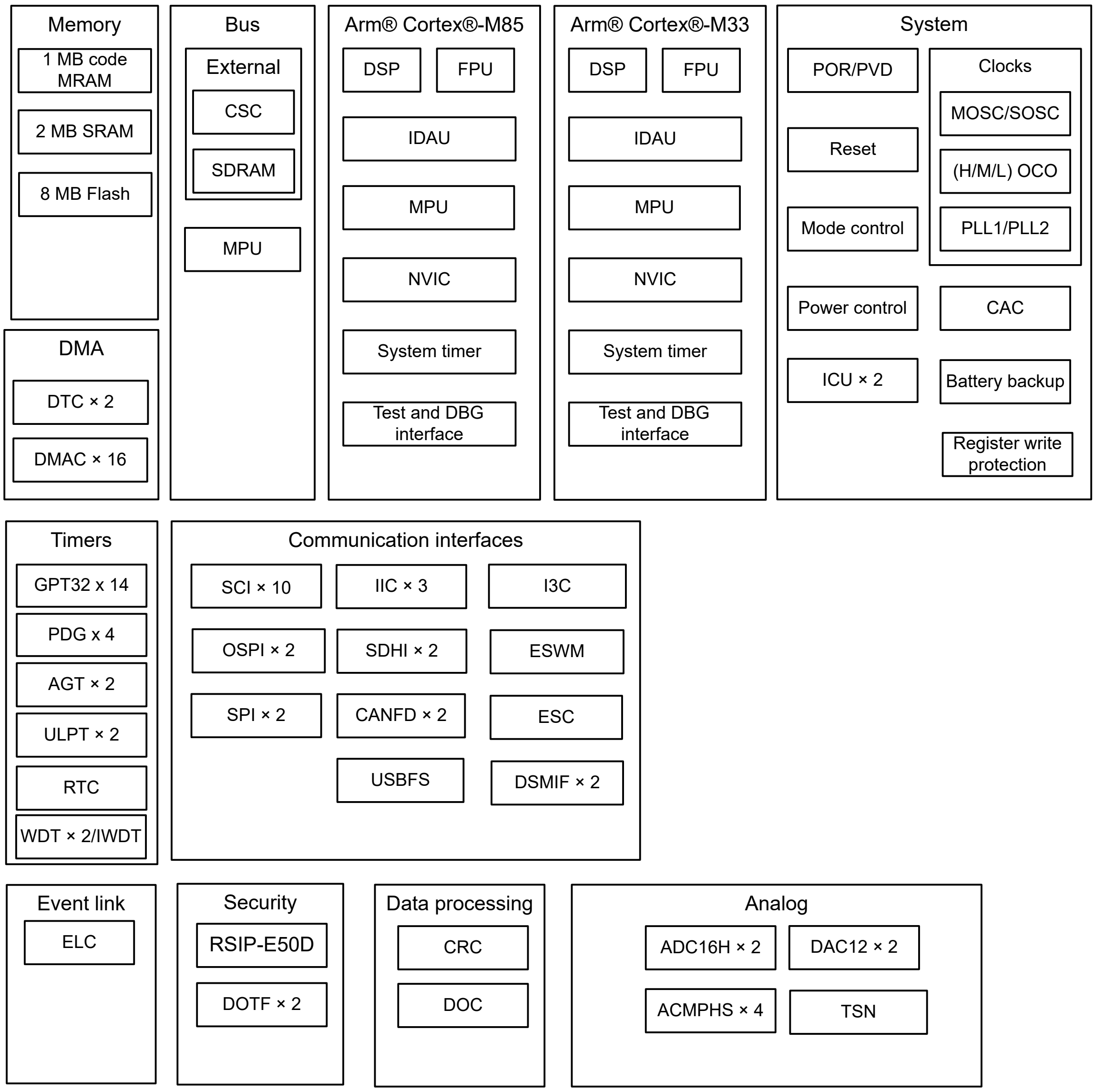
| Feature | Functional description |
|---|---|
| Data Operation Circuit (DOC) | The Data Operation Circuit (DOC) compares, adds, and subtracts 32-bits data. When a selected condition applies, 32-bit data is compared and an interrupt can be generated.<br>See section 56, Data Operation Circuit (DOC). |

**Table 1.11 Security**

| Feature | Functional description |
|---|---|
| Security function | • ARMv8-M TrustZone security<br>• Privileged control<br>• Device lifecycle management<br>• Authentication Level (AL)<br>• Key injection<br>• Secure pin multiplexing<br>• HUK zeroization<br>• VBATT backup registers zeroization<br>• Secure boot<br>• Secure factory programming<br>See section 50, Security Features. |
| Renesas Secure IP (RSIP-E50D) | • Symmetric cryptography: AES and ChaCha20-Poly1305<br>• Asymmetric cryptography: RSA and ECC<br>• Message digest computation: HASH, HMAC<br>• 128-bit true random number generation circuit<br>• 256-bit Hardware Unique Key (HUK)<br>• 128-bit unique ID<br>• OEM boot loader version<br>• Key data for the decryption on-the-fly (DOTF)<br>• SPA/DPA Protections<br>See section 51, Renesas Secure IP (RSIP-E50D). |
| Decryption on-the-fly (DOTF) | Decryption on-the-fly (DOTF) decrypts the encrypted content stored in the external memory in real-time.<br>See section 45, Decryption On The Fly (DOTF). |

## 1.2 Block Diagram

Figure 1.1 shows a block diagram of the MCU superset. Some individual devices within the group have a subset of the features.




Note: Not available on all part numbers.

**Figure 1.1 Block diagram**

## 1.3 Part Numbering

Figure 1.2 shows the product part number information, including memory capacity and package type. Table 1.12 shows a list of products.

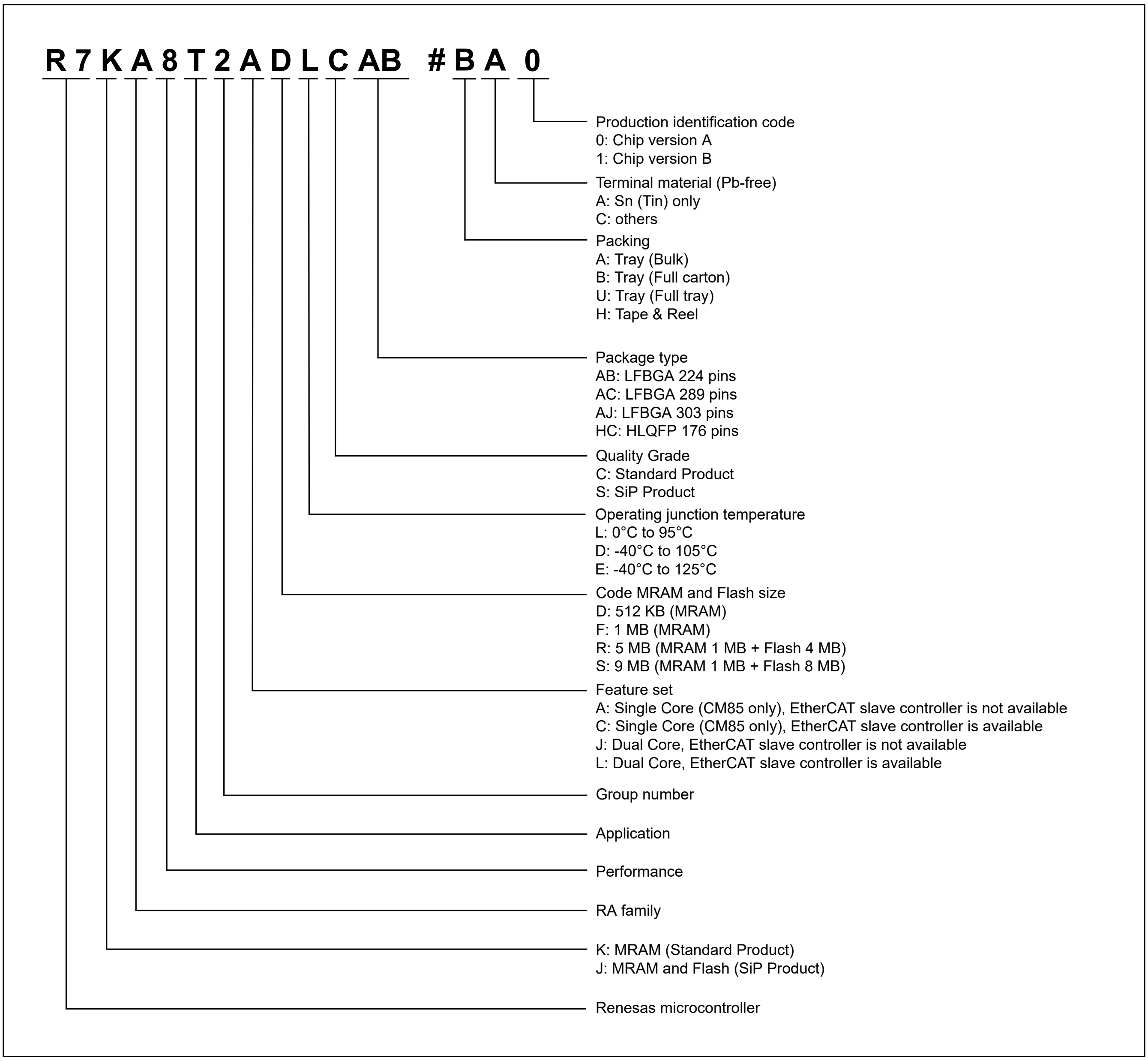



**Figure 1.2 Part numbering scheme**

**Table 1.12 Product list**

| Product part number | Product group | CPU | EtherCAT | Package code | Code MRAM | SRAM | Flash | Operating junction temperature |
|---|---|---|---|---|---|---|---|---|
| R7KA8T2ADDCAB | B | single | — | PLBG0224JA-A | 512 KB | 2 MB | — | -40 to 105 °C |
| R7KA8T2ADECAB | C | single | — | PLBG0224JA-A | 512 KB | 2 MB | — | -40 to 125 °C |
| R7KA8T2ADECHC | C | single | — | PLQP0176KK-A | 512 KB | 2 MB | — | -40 to 125 °C |
| R7KA8T2AFLCAB | A | single | — | PLBG0224JA-A | 1 MB | 2 MB | — | 0 to 95 °C |
| R7KA8T2AFLCAC | A | single | — | PLBG0289JA-A | 1 MB | 2 MB | — | 0 to 95 °C |
| R7KA8T2AFDCAB | B | single | — | PLBG0224JA-A | 1 MB | 2 MB | — | -40 to 105 °C |
| R7KA8T2AFDCAC | B | single | — | PLBG0289JA-A | 1 MB | 2 MB | — | -40 to 105 °C |
| R7KA8T2AFECAB | C | single | — | PLBG0224JA-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2AFECAC | C | single | — | PLBG0289JA-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2AFECHC | C | single | — | PLQP0176KK-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2CDDCAB | B | single | ✔ | PLBG0224JA-A | 512 KB | 2 MB | — | -40 to 105 °C |
| R7KA8T2CDECAB | C | single | ✔ | PLBG0224JA-A | 512 KB | 2 MB | — | -40 to 125 °C |
| R7KA8T2CDECHC | C | single | ✔ | PLQP0176KK-A | 512 KB | 2 MB | — | -40 to 125 °C |
| R7KA8T2CFLCAB | A | single | ✔ | PLBG0224JA-A | 1 MB | 2 MB | — | 0 to 95 °C |
| R7KA8T2CFLCAC | A | single | ✔ | PLBG0289JA-A | 1 MB | 2 MB | — | 0 to 95 °C |
| R7KA8T2CFDCAB | B | single | ✔ | PLBG0224JA-A | 1 MB | 2 MB | — | -40 to 105 °C |
| R7KA8T2CFDCAC | B | single | ✔ | PLBG0289JA-A | 1 MB | 2 MB | — | -40 to 105 °C |
| R7KA8T2CFECAB | C | single | ✔ | PLBG0224JA-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2CFECAC | C | single | ✔ | PLBG0289JA-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2CFECHC | C | single | ✔ | PLQP0176KK-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2JFLCAB | A | dual | — | PLBG0224JA-A | 1 MB | 2 MB | — | 0 to 95 °C |
| R7KA8T2JFLCAC | A | dual | — | PLBG0289JA-A | 1 MB | 2 MB | — | 0 to 95 °C |
| R7KA8T2JFDCAB | B | dual | — | PLBG0224JA-A | 1 MB | 2 MB | — | -40 to 105 °C |
| R7KA8T2JFDCAC | B | dual | — | PLBG0289JA-A | 1 MB | 2 MB | — | -40 to 105 °C |
| R7KA8T2JFECAB | C | dual | — | PLBG0224JA-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2JFECAC | C | dual | — | PLBG0289JA-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2JFECHC | C | dual | — | PLQP0176KK-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2LFLCAB | A | dual | ✔ | PLBG0224JA-A | 1 MB | 2 MB | — | 0 to 95 °C |
| R7KA8T2LFLCAC | A | dual | ✔ | PLBG0289JA-A | 1 MB | 2 MB | — | 0 to 95 °C |
| R7KA8T2LFDCAB | B | dual | ✔ | PLBG0224JA-A | 1 MB | 2 MB | — | -40 to 105 °C |
| R7KA8T2LFDCAC | B | dual | ✔ | PLBG0289JA-A | 1 MB | 2 MB | — | -40 to 105 °C |
| R7KA8T2LFECAB | C | dual | ✔ | PLBG0224JA-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2LFECAC | C | dual | ✔ | PLBG0289JA-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7KA8T2LFECHC | C | dual | ✔ | PLQP0176KK-A | 1 MB | 2 MB | — | -40 to 125 °C |
| R7JA8T2JRLSAJ | A | dual | — | PLBG0303GA-A | 1 MB | 2 MB | 4 MB | 0 to 95 °C |
| R7JA8T2JSLSAJ | A | dual | — | PLBG0303GA-A | 1 MB | 2 MB | 8 MB | 0 to 95 °C |
| R7JA8T2JRDSAJ | B | dual | — | PLBG0303GA-A | 1 MB | 2 MB | 4 MB | -40 to 105 °C |
| R7JA8T2JSDSAJ | B | dual | — | PLBG0303GA-A | 1 MB | 2 MB | 8 MB | -40 to 105 °C |
| R7JA8T2LRLSAJ | A | dual | ✔ | PLBG0303GA-A | 1 MB | 2 MB | 4 MB | 0 to 95 °C |
| R7JA8T2LSLSAJ | A | dual | ✔ | PLBG0303GA-A | 1 MB | 2 MB | 8 MB | 0 to 95 °C |
| R7JA8T2LRDSAJ | B | dual | ✔ | PLBG0303GA-A | 1 MB | 2 MB | 4 MB | -40 to 105 °C |
| R7JA8T2LSDSAJ | B | dual | ✔ | PLBG0303GA-A | 1 MB | 2 MB | 8 MB | -40 to 105 °C |

## 1.4 Function Comparison

**Table 1.13 Function Comparison (1 of 2)**

| Parts number | | R7KA 8T2Ax xCAC | R7KA 8T2Cx xCAC | R7KA 8T2Jx xCAC | R7KA 8T2Lx xCAC | R7KA 8T2Ax xCAB | R7KA 8T2Cx xCAB | R7KA 8T2Jx xCAB | R7KA 8T2Lx xCAB | R7KA 8T2Ax xCHC | R7KA 8T2Cx xCHC | R7KA 8T2Jx xCHC | R7KA 8T2Lx xCHC | R7JA 8T2Jx xSAJ | R7JA 8T2Lx xSAJ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Pin count | | 289 | | | | 224 | | | | 176 | | | | 303 | |
| Package | | BGA | | | | | | | | HLQFP | | | | BGA | |
| I/O Port | | 215 | | | | 156 | | | | 137 | | | | 202 | |
| Code MRAM | | 1 MB, 512 KB | | | | | | | | | | | | 1 MB | |
| CPU0 TCM | | 256 KB | | | | | | | | | | | | | |
| CPU1 TCM | | No | | 128 KB | | No | | 128 KB | | No | | 128 KB | | | |
| CPU0 I/D Caches | | 32 KB | | | | | | | | | | | | | |
| CPU1 C/S Caches | | No | | 32 KB | | No | | 32 KB | | No | | 32 KB | | | |
| SRAM | | 1792 KB | | 1664 KB | | 1792 KB | | 1664 KB | | 1792 KB | | 1664 KB | | | |
| Flash | | No | | | | | | | | | | | | 8 MB, 4 MB | |
| DMA | DTC | 1 | | 2 | | 1 | | 2 | | 1 | | 2 | | | |
| | DMAC | 8 | | 16 | | 8 | | 16 | | 8 | | 16 | | | |
| BUS | External bus | 32-bit bus | | | | 16-bit bus | | | | | | | | | |
| | SDRAM | 32-bit bus | | | | 16-bit bus | | | | | | | | | |
| System | CPU0 clock | 1 GHz (max.) | | | | | | | | 600 MHz (max.) | | | | 1 GHz (max.) | |
| | CPU1 clock | No | | 250 MHz (max.) | | No | | 250 MHz (max.) | | No | | 200 MHz (max.) | | 250 MHz (max.) | |
| | CPUs clock sources | MOSC, SOSC, HOCO, MOCO, PLL1P | | | | | | | | | | | | | |
| | CAC | Yes | | | | | | | | | | | | | |
| | WDT | 1 | | 2 | | 1 | | 2 | | 1 | | 2 | | | |
| | IWDT | Yes | | | | | | | | | | | | | |
| | Backup register | 128 B | | | | | | | | | | | | | |
| Communication | SCI | 10 | | | | 9 | | | | | | | | 10 | |
| | IIC | 3 | | | | | | | | | | | | | |
| | I3C | Yes | | | | | | | | | | | | | |
| | SPI | 2 | | | | | | | | | | | | | |
| | CANFD | 2 | | | | | | | | | | | | | |
| | USBFS | Yes | | | | | | | | | | | | | |
| | OSPI | 2 | | | | 1 | | | | | | | | 2*2 | |
| | SDHI/ MMC | 2 | | | | | | | | | | | | | |
| | ESWM | MII, RMII, GMII, RGMII | | | | MII, RMII, RGMII | | | | MII, RMII | | | | MII, RMII, GMII, RGMII | |
| | ESC | No | Yes | No | Yes | No | Yes | No | Yes | No | Yes | No | Yes | No | Yes |
| | DSMIF | 2 | | | | | | | | | | | | | |

**Table 1.13 Function Comparison (2 of 2)**

| Parts number | | R7KA 8T2Ax xCAC | R7KA 8T2Cx xCAC | R7KA 8T2Jx xCAC | R7KA 8T2Lx xCAC | R7KA 8T2Ax xCAB | R7KA 8T2Cx xCAB | R7KA 8T2Jx xCAB | R7KA 8T2Lx xCAB | R7KA 8T2Ax xCHC | R7KA 8T2Cx xCHC | R7KA 8T2Jx xCHC | R7KA 8T2Lx xCHC | R7JA 8T2Jx xSAJ | R7JA 8T2Lx xSAJ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Timers | GPT32 *1 | 14 | | | | | | | | | | | | | |
| | PDG | 4 | | | | | | | | | | | | | |
| | AGT*1 | 2 | | | | | | | | | | | | | |
| | ULPT*1 | 2 | | | | | | | | | | | | | |
| | RTC | Yes | | | | | | | | | | | | | |
| Analog | ADC16H | Unit 0: 15, Unit 1: 15 | | | | Unit 0: 7, Unit 1: 5 | | | | | | | | Unit 0: 15, Unit 1: 15 | |
| | DAC12 | 2 | | | | | | | | | | | | | |
| | ACMPHS | 4 | | | | | | | | | | | | | |
| | TSN | Yes | | | | | | | | | | | | | |
| Data processing | CRC | Yes | | | | | | | | | | | | | |
| | DOC | Yes | | | | | | | | | | | | | |
| Event control | ELC | Yes | | | | | | | | | | | | | |
| Security | | RSIP-E50D, Decryption on-the-fly, Secure Debug, OTP, TrustZone, and Lifecycle management | | | | | | | | | | | | | |

Note: The product name differs depending on the supported memory size. See section 1.3. Part Numbering.
Note: The maximum frequency varies by product group. See section 60.3.1. Frequency for details.
Note 1. Available pins depend on the Pin count, about details see section 1.7. Pin Lists.
Note 2. OSPI1 is connected to the serial Flash in the SiP product.

## 1.5 Pin Functions

**Table 1.14 Pin functions (1 of 8)**

| Function | Signal | I/O | Description |
|---|---|---|---|
| Power supply | VCC_01 to VCC_10, VCC2_11 to VCC2_15 | Input | Power supply pin. Connect it to the system power supply. Connect this pin to the same numbered VSS_01 to VSS_15 by a 0.1-μF capacitor. The capacitor should be placed close to the pin.<br>In the SiP product, connect VCC2_11 to VCC2_15 to the 1.8V system power supply. |
| | VCC2_16 to VCC2_19 | Input | Dedicated power supply pin for the SiP product. Connect it to the 1.8V system power supply. Connect this pin to the same numbered VSS_16 to VSS_19 by a 0.1-μF capacitor. The capacitor should be placed close to the pin. |
| | VCC_DCDC | Input | Switching regulator power supply pin. |
| | VLO | I/O | Switching regulator pin. |
| | VCL0 to VCL11 | Input | Connect this pin to the same numbered VSS0 to VSS11 pin by the smoothing capacitor used to stabilize the internal power supply.<br>Place the capacitor close to the pin. |
| | VBATT | Input | Battery Backup power pin. |
| | VSS_01 to VSS_15, VSS0 to VSS11, VSS_DCDC | Input | Ground pin. Connect it to the system power supply (0 V). |
| | VSS_16 to VSS_19, VSS | Input | Dedicated ground pin for the SiP product. Connect it to the system power supply (0 V). |
| | Vpp | Input | Power supply pin for serial Flash programming operation. See the ISSI serial Flash IS25WX064 datasheet for the details. If not used, the Vpp pin can be floated. Note that the deep power down current varies depending on the power supply status of the Vpp pin. |
| Clock | XTAL | Output | Pins for a crystal resonator. An external clock signal can be input through the EXTAL pin. |
| | EXTAL | Input | |
| | XCIN | Input | Input/output pins for the sub-clock oscillator. Connect a crystal resonator between XCOUT and XCIN. |
| | XCOUT | Output | |
| | EXCIN | Input | External sub-clock input |
| | CLKOUT | Output | Clock output pin |
| Operating mode control | MD | Input | Pin for setting the operating mode. The signal level on this pin must not be changed during operation mode transition on release from the reset state. |
| System control | RES | Input | Reset signal input pin. The MCU enters the reset state when this signal goes low. |
| | PUP | Input | Connect to VCC2 through a resistor. |
| CAC | CACREF | Input | Measurement reference clock input pin |
| On-chip emulator | TMS | Input | On-chip emulator or boundary scan pins |
| | TDI | Input | |
| | TCK | Input | |
| | TDO | Output | |
| | TCLK | Output | Output clock for synchronization with the trace data |
| | TDATA0 to TDATA3 | Output | Trace data output |
| | SWO | Output | Serial wire trace output pin |
| | SWDIO | I/O | Serial wire debug data input/output pin |
| | SWCLK | Input | Serial wire clock pin |

**Table 1.14 Pin functions (2 of 8)**

| Function | Signal | I/O | Description |
|---|---|---|---|
| Interrupt | NMI | Input | Non-maskable interrupt request pin |
| | IRQn | Input | Maskable interrupt request pins |
| | IRQn-DS | Input | Maskable interrupt request pins that can also be used in Deep Software Standby mode |
| External bus interface | EBCLK | Output | Outputs the external bus clock for external devices |
| | RD | Output | Strobe signal indicating that reading from the external bus interface space is in progress, active-low |
| | WR | Output | Strobe signal indicating that writing to the external bus interface space is in progress, in 1-write strobe mode, active-low |
| | WRn | Output | Strobe signals indicating that either group of data bus pins (D07 to D00, D15 to D08, D23 to D16 or D31 to D24) is valid in writing to the external bus interface space, in byte strobe mode, active-low |
| | BCn | Output | Strobe signals indicating that either group of data bus pins (D07 to D00, D15 to D08, D23 to D16 or D31 to D24) is valid in access to the external bus interface space, in 1-write strobe mode, active-low |
| | ALE | Output | Address latch signal when address/data multiplexed bus is selected |
| | WAIT | Input | Input pin for wait request signals in access to the external space, active-low |
| | CSn | Output | Select signals for CS areas, active-low |
| | A00 to A23 | Output | Address bus |
| | D00 to D31 | I/O | Data bus |
| | A00/D00 to A15/D15 | I/O | Address/data multiplexed bus |
| SDRAM interface | SDCLK | Output | Outputs the SDRAM-dedicated clock |
| | CKE | Output | SDRAM clock enable signal |
| | SDCS | Output | SDRAM chip select signal, active low |
| | RAS | Output | SDRAM low address strobe signal, active low |
| | CAS | Output | SDRAM column address strobe signal, active low |
| | WE | Output | SDRAM write enable signal, active low |
| | DQMn | Output | SDRAM I/O data mask enable signal for DQ07 to DQ00, DQ15 to DQ08, DQ23 to DQ16 or DQ31 to DQ24 |
| | A00 to A16 | Output | Address bus |
| | DQ00 to DQ31 | I/O | Data bus |

**Table 1.14 Pin functions (3 of 8)**

| Function | Signal | I/O | Description |
|---|---|---|---|
| GPT | GTETRGA, GTETRGB, GTETRGC, GTETRGD | Input | External trigger input pins |
| | GTIOCnA, GTIOCnB | I/O | Input capture, output compare, or PWM output pins |
| | GTADSM0, GTADSM1 | Output | A/D conversion start request monitoring output pins |
| | GTCPPOn | Output | Toggle output synchronized with PWM period |
| | GTIU | Input | Hall sensor input pin U |
| | GTIV | Input | Hall sensor input pin V |
| | GTIW | Input | Hall sensor input pin W |
| | GTOUUP | Output | 3-phase PWM output for BLDC motor control (positive U phase) |
| | GTOULO | Output | 3-phase PWM output for BLDC motor control (negative U phase) |
| | GTOVUP | Output | 3-phase PWM output for BLDC motor control (positive V phase) |
| | GTOVLO | Output | 3-phase PWM output for BLDC motor control (negative V phase) |
| | GTOWUP | Output | 3-phase PWM output for BLDC motor control (positive W phase) |
| | GTOWLO | Output | 3-phase PWM output for BLDC motor control (negative W phase) |
| AGT | AGTEEn | Input | External event input enable signals |
| | AGTIOn | I/O | External event input and pulse output pins |
| | AGTOn | Output | Pulse output pins |
| | AGTOAn | Output | Output compare match A output pins |
| | AGTOBn | Output | Output compare match B output pins |
| ULPT | ULPTEEn | Input | External count control input |
| | ULPTEVIn | Input | External event input |
| | ULPTEEn-DS | Input | External count control input that can also be used in Deep Software Standby mode 1 |
| | ULPTEVIn-DS | Input | External event input that can also be used in Deep Software Standby mode 1 |
| | ULPTOn | Output | Pulse output |
| | ULPTOAn | Output | Output compare match A output |
| | ULPTOBn | Output | Output compare match B output |
| | ULPTOn-DS | Output | Pulse output that can also be used in Deep Software Standby mode 1 |
| | ULPTOAn-DS | Output | Output compare match A output that can also be used in Deep Software Standby mode 1 |
| | ULPTOBn-DS | Output | Output compare match B output that can also be used in Deep Software Standby mode 1 |
| RTC | RTCOUT | Output | Output pin for 1-Hz or 64-Hz clock |
| | RTCICn | Input | Time capture event input pins |

**Table 1.14 Pin functions (4 of 8)**

| Function | Signal | I/O | Description |
|---|---|---|---|
| SCI | SCKn | I/O | Input/output pins for the clock (clock synchronous mode) |
| | RXDn | Input | Input pins for received data (asynchronous mode/clock synchronous mode) |
| | TXDn | Output | Output pins for transmitted data (asynchronous mode/clock synchronous mode) |
| | CTSn_RTSn | I/O | Input/output pins for controlling the start of transmission and reception (asynchronous mode/clock synchronous mode), active-low. |
| | CTSn | Input | Input for the start of transmission. |
| | DEn | Output | Driver enable signal for RS-485 |
| | SCLn | I/O | Input/output pins for the IIC clock (simple IIC mode) |
| | SDAn | I/O | Input/output pins for the IIC data (simple IIC mode) |
| | SCKn | I/O | Input/output pins for the clock (simple SPI mode) |
| | MISOn | I/O | Input/output pins for slave transmission of data (simple SPI mode) |
| | MOSIn | I/O | Input/output pins for master transmission of data (simple SPI mode) |
| | SSn | Input | Chip-select input pins (simple SPI mode), active-low |
| IIC | SCLn | I/O | Input/output pins for the clock |
| | SDAn | I/O | Input/output pins for data |
| I3C | I3C_SCL0 | I/O | Input/output pins for the clock |
| | I3C_SDA0 | I/O | Input/output pins for data |
| SPI | RSPCKA, RSPCKB | I/O | Clock input/output pin |
| | MOSIA, MOSIB | I/O | Input or output pins for data output from the master |
| | MISOA, MISOB | I/O | Input or output pins for data output from the slave |
| | SSLA0, SSLB0 | I/O | Input or output pin for slave selection |
| | SSLA1 to SSLA3, SSLB1 to SSLB3 | Output | Output pins for slave selection |
| CANFD | CRXn | Input | Receive data |
| | CTXn | Output | Transmit data |
| USBFS | VCC_USB | Input | Power supply pin |
| | VSS_USB | Input | Ground pin |
| | USB_DP | I/O | D+ pin of the USB on-chip transceiver. Connect this pin to the D+ pin of the USB bus. |
| | USB_DM | I/O | D- pin of the USB on-chip transceiver. Connect this pin to the D- pin of the USB bus. |
| | USB_VBUS | Input | USB cable connection monitor pin. Connect this pin to VBUS of the USB bus. The VBUS pin status (connected or disconnected) can be detected when the USB module is operating as a function controller. |
| | USB_EXICEN | Output | Low-power control signal for external power supply (OTG) chip |
| | USB_VBUSEN | Output | VBUS (5 V) supply enable signal for external power supply chip |
| | USB_OVRCURA, USB_OVRCURB | Input | Connect the external overcurrent detection signals to these pins. Connect the VBUS comparator signals to these pins when the OTG power supply chip is connected. |
| | USB_OVRCURA-DS, USB_OVRCURB-DS | Input | Overcurrent pins for USBFS that can also be used in Deep Software Standby mode 1. Connect the external overcurrent detection signals to these pins. Connect the VBUS comparator signals to these pins when the OTG power supply chip is connected. |
| | USB_ID | Input | Connect the MicroAB connector ID input signal to this pin during operation in OTG mode |

**Table 1.14 Pin functions (5 of 8)**

| Function | Signal | I/O | Description |
|---|---|---|---|
| OSPI | OM_n_SCLK | Output | Clock output (OCTACLK divided by 2) |
| | OM_n_SCLKN | Output | Inverted clock output (OCTACLK divided by 2) |
| | OM_n_CSn | Output | Chip select signal for an OctaFlash device, active-low |
| | OM_n_DQS | I/O | Read data strobe/write data mask signal |
| | OM_n_SIOn | I/O | Data input/output |
| | OM_n_RESET | Output | Reset signal for both slave devices, active-low |
| | OM_n_ECSINT1 | Input | Error Correction Status and Interrupt for slave1 |
| | OM_n_RSTO1 | Input | Slave reset status for slave1 |
| | OM_n_WP1 | Output | Write Protect for slave1, active-low |
| SDHI/MMC | SDnCLK | Output | SD clock output pins |
| | SDnCMD | I/O | Command output pin and response input signal pins |
| | SDnDAT0 to SDnDAT7 | I/O | SD and MMC data bus pins |
| | SDnCD | Input | SD card detection pins |
| | SDnWP | Input | SD write-protect signals |

**Table 1.14 Pin functions (6 of 8)**

| Function | Signal | I/O | Description |
|---|---|---|---|
| ESWM | ETn_GTX_CLK | Output | 1000 Mb/s transmit clock |
| | ETn_TX_CLK | Input | 100 Mb/s,10 Mb/s transmit clock |
| | ETn_RX_CLK | Input | Receive clock |
| | ETn_TX_EN | Output | Transmit enable |
| | ETn_TXD0 to ETn_TXD7 | Output | Transmit data |
| | ETn_TX_ER | Output | Transmit coding error |
| | ETn_RX_DV | Input | Receive data valid |
| | ETn_RXD0 to ETn_RXD7 | Input | Receive data |
| | ETn_RX_ER | Input | Receive error |
| | ETn_MDC | Output | Management data clock |
| | ETn_MDIO | I/O | Management data input/output |
| | RGMIIn_TXC | Output | Transmit clock |
| | RGMIIn_RXC | Input | Receive clock |
| | RGMIIn_TX_CTL | Output | Transmit control |
| | RGMIIn_TXD0 to RGMIIn_TXD3 | Output | Transmit data |
| | RGMIIn_RX_CTL | Input | Receive control |
| | RGMIIn_RXD0 to RGMIIn_RXD3 | Input | Receive data |
| | RMIIn_REF50CK | Input | Synchronous clock reference |
| | RMIIn_TX_EN | Output | Transmit enable |
| | RMIIn_TXD0 to RMIIn_TXD1 | Output | Transmit data |
| | RMIIn_CRS_DV | Input | Carrier sense/Receive data valid |
| | RMIIn_RXD0 to RMIIn_RXD1 | Input | Receive data |
| | RMIIn_RX_ER | Input | Receive error |
| | ETn_LINKSTA | Input | PHY Link Status |
| | ETn_INT | Input | PHY interrupt |
| | ETn_WOL | Output | Wake-on-LAN. This signal indicates that a Magic Packet was received. |
| | GPTP_CAPTUREn | Input | Media clock capture input |
| | GPTP_MATCHn | Output | Media clock recovery output |
| | GPTP_PPSn | Output | PPS signal |
| | GPTP_PTPOUT0 to GPTP_PTPOUT3 | Output | PTP Pulse generator signal |
| | ET_TAS_STA0 to ET_TAS_STA3 | Output | TAS status monitor |
| | ETHPHYCLK | Output | Clock output for PHY (Shared with ESC) |

**Table 1.14 Pin functions (7 of 8)**

| Function | Signal | I/O | Description |
|---|---|---|---|
| ESC | CATn_LINKSTA | Input | Link status inputs from the PHY-LSI |
| | CATn_RX_CLK | Input | Receive clocks |
| | CATn_RX_DV | Input | Receive data valid |
| | CATn_ERXD0 to CATn_ERXD3 | Input | Receive data |
| | CATn_RX_ER | Input | Receive error |
| | CATn_TX_CLK | Input | Transmit clocks |
| | CATn_TX_EN | Output | Transmit enable |
| | CATn_ETXD0 to CATn_ETXD3 | Output | Transmit data |
| | CAT0_MDC | Output | Management data clock |
| | CAT0_MDIO | I/O | Management data I/O |
| | ETHPHYCLK | Output | Clock output for PHY (Shared with ESWM) |
| | CATRESETOUT | Output | PHY reset signal |
| | CATLEDRUN | Output | RUN LED (green) output |
| | CATIRQ | Output | IRQ output |
| | CATLEDSTER | Output | Output for RUN LED part of STATE LED (bicolor) (turned off while ERR) |
| | CATLEDERR | Output | ERR LED (red) output |
| | CATLINKACT0, CATLINKACT1 | Output | Link/Activity LED outputs |
| | CATSYNC0, CATSYNC1 | Output | SYNC signal outputs |
| | CATLATCH0, CATLATCH1 | input | LATCH signal inputs |
| | CATI2CCLK | Output | EEPROM I2C clock output |
| | CATI2CDATA | I/O | EEPROM I2C data |
| DSMIF | DSMnCLK0 to DSMnCLK2 | I/O | Clock input/output pin |
| | DSMnDAT0 to DSMnDAT2 | Input | Data input pin |
| Analog power supply | AVCC0 | Input | Analog voltage supply pin. This is used as the analog power supply for the respective modules. |
| | AVSS0 | Input | Analog ground pin. This is used as the analog ground for the respective modules. Supply this pin with the same voltage as the VSS pin. |
| | VREFH | Input | Analog reference voltage supply pin for the ADC16H (unit 1) and D/A Converter. Connect this pin to AVCC0 when not using the ADC16H (unit 1) and D/A Converter. |
| | VREFL | Input | Analog reference ground pin for the ADC16H and D/A Converter. Connect this pin to AVSS0 when not using the ADC16H (unit 1) and D/A Converter. |
| | VREFH0 | Input | Analog reference voltage supply pin for the ADC16H (unit 0). Connect this pin to AVCC0 when not using the ADC16H (unit 0). |
| | VREFL0 | Input | Analog reference ground pin for the ADC16H. Connect this pin to AVSS0 when not using the ADC16H (unit 0). |

**Table 1.14 Pin functions (8 of 8)**

| Function | Signal | I/O | Description |
|---|---|---|---|
| ADC16H | ANxxx | Input | Input pins for the analog signals to be processed by the A/D converter. |
| | ADTRGm | Input | Input pins for the external trigger signals that start the A/D conversion, active-low. |
| | ADSTm | Output | AD conversion start |
| | ADmFLAG1 | Output | AD conversion end |
| | ADSYNC | Output | Synchronization signal between units |
| DAC12 | DAn | Output | Output pins for the analog signals processed by the D/A converter. |
| ACMPHS | VCOUT | Output | Comparator output pin |
| | IVREFn | Input | Reference voltage input pins for comparator |
| | IVCMPn | Input | Analog voltage input pins for comparator |
| I/O ports | Pmn | I/O | General-purpose input/output pins (m: port number, n: pin number) |
| | P200 | Input | General-purpose input pin |

## 1.6 Pin Assignments

The following figures show the pin assignments from the top view.

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | P609 | P113 | P115 | P112 | P302 | P915 | VLO | VLO | VSS_DCDC | VCC_DCDC | VCC_DCDC | P309 | P906 | P905 | P907 | P904 | P207 | A |
| B | P813 | PA12 | P114 | PA11 | P300 | P303 | VLO | VLO | VSS_DCDC | VCC_DCDC | VCC_DCDC | P311 | P908 | P909 | P206 | PD01 | PD02 | B |
| C | PA06 | P613 | PA13 | P301 | P200 | P210/TMS/SWDIO | P208/TDI | P110 | P308 | P305 | P307 | P911 | P312 | PD04 | PD03 | PD05 | PD06 | C |
| D | PA04 | P611 | P610 | PA14 | RES | P211/TCK/SWCLK | P109 | P108 | P903 | P304 | P306 | P912 | PB04 | PB07 | PB05 | PB03 | PB01 | D |
| E | PA15 | P615 | P614 | P612 | P914 | P201/MD | P209/TDO | P111 | P902 | P310 | P910 | P913 | PB02 | PB06 | PD07 | PB00 | P706 | E |
| F | PA02 | PA10 | PA08 | PA09 | PC14 | VCC_08 | VSS_08 | VSS3 | VCL3 | VSS_07 | VCC_07 | P700 | P702 | P406 | P701 | P707 | P705 | F |
| G | PA00 | PA03 | PA05 | PA07 | PC12 | VCC_09 | VSS_09 | VSS4 | VCL4 | VSS_06 | VCC_06 | P405 | P704 | P703 | PB14 | PB12 | PB11 | G |
| H | P504 | P503 | P505 | PA01 | PC11 | VCC_10 | VSS_10 | VSS7 | VCL5 | VSS5 | VCC_04 | VSS_04 | P403 | VCC_05 | PB09 | PB10 | PB13 | H |
| J | P506 | P507 | P508 | P509 | PC13 | VCC2_11 | VSS_11 | VCL7 | VCL6 | VSS6 | VCL2 | VSS2 | P404 | PB08 | VSS_05 | VSS_03 | VSS_02 | J |
| K | PC15 | P608 | P510 | PD00 | PC07 | VSS_12 | VSS9 | VCL9 | VCL8 | VSS8 | VCL1 | VSS1 | P410 | VCC_03 | VCC_02 | P213/XTAL | P212/EXTAL | K |
| L | PC03 | PC02 | PC04 | PC09 | PC05 | VCC2_12 | VSS_14 | VSS_15 | VSS10 | VCL10 | VCL0 | VSS0 | P414 | P402 | VCC_01 | P214/XCOUT | P215/XCIN/EXCIN | L |
| M | PC00 | P607 | PC01 | PC08 | PC10 | P104 | VCC2_14 | VCC2_15 | P810 | VSS11 | VCL11 | P412 | P710 | P411 | P408 | VBATT | VSS_01 | M |
| N | P605 | P604 | P606 | PC06 | P107 | P106 | P105 | P811 | P013 | P011 | P807 | P708 | P712 | P714 | P711 | P713 | P401 | N |
| P | P603 | P602 | P600 | P601 | P102 | P801 | P803 | P812 | P012 | P010 | P009 | P805 | P512 | P413 | P515 | P709 | P400 | P |
| R | VCC2_13 | P315 | P900 | P103 | P101 | P802 | P804 | P501 | AVCC0 | AVSS0 | P005 | P003 | P513 | P514 | P415 | P409 | P407 | R |
| T | P205 | P203 | P313 | P901 | P809 | P800 | P502 | P014 | VREFL | VREFL0 | P004 | P007 | P001 | P806 | P715 | P815/USB_DM | VSS_USB | T |
| U | P204 | P202 | P314 | VSS_13 | P808 | P100 | P500 | P015 | VREFH | VREFH0 | P008 | P006 | P000 | P002 | P511 | P814/USB_DP | VCC_USB | U |
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | |

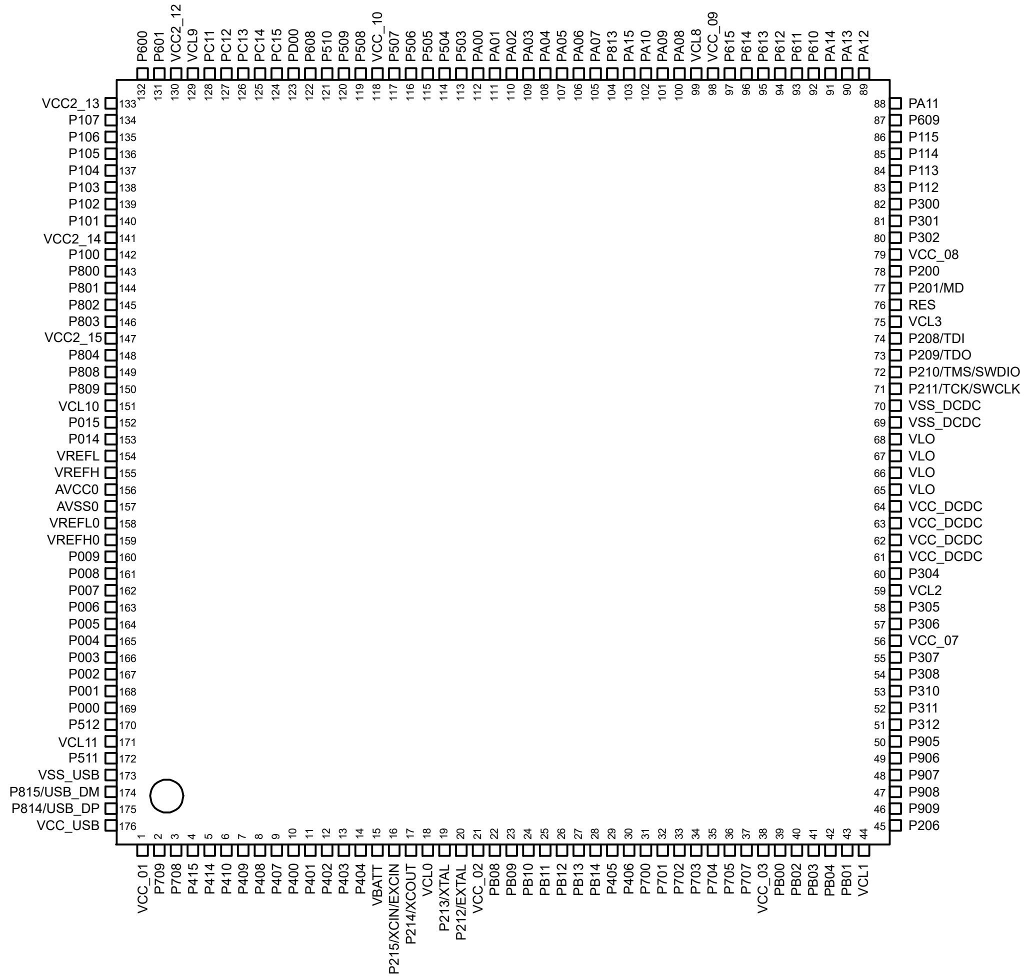
**Figure 1.3 Pin assignment for BGA 289-pin**

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | NC | PA11 | P114 | P112 | P300 | VLO | VLO | VSS_DCDC | VCC_DCDC | VCC_DCDC | P309 | P312 | P908 | P905 | P206 | A |
| B | P610 | PA12 | P115 | P113 | P302 | VLO | VLO | VSS_DCDC | VCC_DCDC | VCC_DCDC | P311 | P310 | P906 | P907 | P909 | B |
| C | P612 | P611 | PA13 | P609 | P301 | RES | P210/TMS/SWDIO | P211/TCK/SWCLK | P304 | P306 | P305 | P307 | PB03 | PB00 | PB01 | C |
| D | P615 | P613 | P614 | PA14 | P200 | P208/TDI | P201/MD | P209/TDO | P902 | P308 | PB02 | PB04 | P705 | P707 | P706 | D |
| E | PA15 | PA08 | P813 | PA09 | VCC_08 | VSS_08 | VSS5 | VCL5 | VSS_07 | VCC_07 | P405 | P702 | P704 | P406 | P701 | E |
| F | PA06 | PA10 | PA05 | PA07 | VCC_09 | VSS_09 | VSS6 | VCL6 | VCL4 | VSS4 | P700 | P703 | PB14 | PB12 | PB11 | F |
| G | PA04 | PA02 | PA01 | PA03 | VCC_10 | VSS_10 | VSS7 | VCL7 | VCL3 | VSS3 | P404 | VCC_05 | PB09 | PB10 | PB13 | G |
| H | PA00 | P504 | P503 | P505 | PC14 | VSS_15 | VSS8 | VCL8 | VCL2 | VSS2 | P403 | PB08 | VSS_05 | VSS_03 | VSS_02 | H |
| J | P506 | P510 | P507 | P508 | PC12 | VCC2_15 | VSS9 | VCL9 | VCL1 | VSS1 | P402 | VCC_03 | VCC_02 | P213/XTAL | P212/EXTAL | J |
| K | PC15 | P608 | PD00 | P509 | VCC2_14 | VSS_14 | VSS10 | VCL10 | VCL0 | VSS0 | P410 | P407 | VCC_01 | P214/XCOUT | P215/XCIN/EXCIN | K |
| L | PC13 | P604 | P603 | P107 | P106 | P104 | P105 | VSS11 | VCL11 | P409 | P414 | P408 | P415 | VBATT | VSS_01 | L |
| M | PC11 | P602 | P600 | P601 | P102 | P801 | P803 | P009 | P007 | P708 | P411 | P710 | P709 | P711 | P401 | M |
| N | VCC2_12 | P315 | VSS_13 | P103 | P101 | P802 | P804 | AVCC0 | AVSS0 | P005 | P001 | P712 | P714 | P713 | P400 | N |
| P | P205 | P203 | P313 | VCC2_13 | P809 | P800 | P015 | VREFL | VREFL0 | P006 | P002 | P003 | P512 | P815/USB_DM | VSS_USB | P |
| R | P204 | P202 | P314 | VSS_12 | P808 | P100 | P014 | VREFH | VREFH0 | P008 | P004 | P000 | P511 | P814/USB_DP | VCC_USB | R |
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | |

**Figure 1.4 Pin assignment for BGA 224-pin**

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | VSS | P114 | P609 | P113 | P301 | P208/TDI | P210/TMS/SWDIO | VLO | VLO | VSS_DCDC | VCC_DCDC | VCC_DCDC | P309 | P906 | P905 | P907 | P207 | VSS | A |
| B | P813 | PA12 | P115 | PA11 | P112 | P209/TDO | P211/TCK/SWCLK | VLO | VLO | VSS_DCDC | VCC_DCDC | VCC_DCDC | P311 | P908 | P909 | P904 | PD01 | PD02 | B |
| C | PA06 | P613 | PA13 | P300 | P302 | P200 | RES | P110 | P903 | P308 | P305 | P307 | P911 | P206 | PD04 | PD03 | PD05 | PD06 | C |
| D | PA04 | P611 | P610 | PA14 | P303 | P915 | P108 | P111 | P109 | P310 | P304 | P306 | P912 | PB04 | PB07 | PB05 | PB03 | PB01 | D |
| E | PA15 | P615 | P614 | P612 | P914 | P201/MD | | | | P902 | P312 | P910 | P913 | PB02 | PB06 | PD07 | PB00 | P706 | E |
| F | PA02 | PA10 | PA08 | PA09 | PC14 | | VCC_08 | VSS_08 | VSS3 | VCL3 | VSS_07 | VCC_07 | P700 | P702 | P406 | P701 | P707 | P705 | F |
| G | PA00 | PA03 | PA05 | PA07 | PC12 | | VCC_09 | VSS_09 | VSS4 | VCL4 | VSS_06 | VCC_06 | P405 | P704 | P703 | PB14 | PB12 | PB11 | G |
| H | P504 | P503 | P505 | PA01 | PC11 | | VCC_10 | VSS_10 | VSS7 | VCL5 | VSS5 | VCC_04 | VSS_04 | | VCC_05 | PB09 | PB10 | PB13 | H |
| J | P506 | P507 | P508 | P509 | PC13 | | VCC2_11 | VSS_11 | VCL7 | VCL6 | VSS6 | VCL2 | VSS2 | | PB08 | VSS_05 | VSS_03 | VSS_02 | J |
| K | PC15 | P608 | P510 | PD00 | VSS | VSS | VSS_12 | VSS9 | VCL9 | VCL8 | VSS8 | VCL1 | VSS1 | | VCC_03 | VCC_02 | P213/XTAL | P212/EXTAL | K |
| L | PC10 | VSS | PUP | VCC2_16 | VSS_16 | | VCC2_12 | VSS_14 | VSS_15 | VSS10 | VCL10 | VCL0 | VSS0 | P403 | P404 | VCC_01 | P214/XCOUT | P215/XCIN/EXCIN | L |
| M | PC09 | VSS | VSS | VCC2_17 | VSS_17 | | | VCC2_14 | VCC2_15 | | VSS11 | VCL11 | | P414 | P402 | P410 | VBATT | VSS_01 | M |
| N | PC08 | VSS | VSS | VCC2_18 | VSS_18 | | P105 | | | P810 | | | | P710 | P411 | P408 | P412 | P401 | N |
| P | VSS | VSS | VSS | VCC2_19 | VSS_19 | P104 | P107 | P106 | P811 | P013 | P011 | P807 | P708 | P712 | P714 | P711 | P713 | P400 | P |
| R | P602 | VSS | VSS | P600 | P601 | P102 | P801 | P803 | P812 | P012 | P010 | P009 | P805 | P512 | P413 | P515 | P709 | P407 | R |
| T | Vpp | VSS | P315 | P900 | P103 | P101 | P802 | P804 | P501 | AVCC0 | AVSS0 | P005 | P003 | P513 | P514 | P415 | P409 | VCC_USB | T |
| U | VCC2_13 | P205 | P203 | P313 | P901 | P809 | P800 | P502 | P014 | VREFL | VREFL0 | P004 | P007 | P001 | P806 | P715 | P815/USB_DM | VSS_USB | U |
| V | VSS | P204 | P202 | P314 | VSS_13 | P808 | P100 | P500 | P015 | VREFH | VREFH0 | P008 | P006 | P000 | P002 | P511 | P814/USB_DP | VSS | V |
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | |

**Figure 1.5 Pin assignment for BGA 303-pin**




Note: VSS_01 to VSS_15 and VSS0 to VSS11 are assigned to Exposed PAD on the QFP package bottom. The PAD must be connected to the ground (0V).

**Figure 1.6 Pin assignment for 176-pin**

## 1.7 Pin Lists

### Table 1.15 Pin list for the Standard product (1 of 8)

| BGA289 | BGA224 | HLQFP176 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/ OSPI/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|---|---|
| A1 | C4 | 87 | — | P609 | D7/DQ7 | IRQ29 | TXD0_C/SDA0_C/MOSI0_C/ MISOA_B/CTX1 | GTIU/GTIOC5B/ULPTOA1-DS | AD1FLAG1 |
| A2 | B4 | 84 | — | P113 | D4/DQ4 | IRQ28 | RXD0_A/SCL0_A/MISO0_A/ SSLA1_B/SD0DAT5_B | GTETRGB/GTIOC2A/ ULPTOA0-DS | ADST1 |
| A3 | B3 | 86 | — | P115 | D6/DQ6 | IRQ31-DS | CTS0_A/MOSIA_B/SD0DAT7_B | GTETRGD/GTIOC5A | AD0FLAG1 |
| A4 | A4 | 83 | — | P112 | D3/DQ3 | IRQ27 | TXD0_A/SDA0_A/MOSI0_A/ SSLA2_B/SD0DAT4_B | GTETRGA/GTIOC3B/ ULPTOB0-DS | ADST0 |
| A5 | B5 | 80 | — | P302 | D0/DQ0 | IRQ5 | RXD6_B/SCL6_B/MISO6_B/ SD0DAT1_B | GTOUUP/GTIOC4A/ ULPTO0-DS | — |
| A6 | — | — | — | P915 | — | IRQ8 | CTS6_B | GTIOC5A | — |
| A7 | A6 | 68 | VLO | — | — | — | — | — | — |
| A8 | A7 | 66 | VLO | — | — | — | — | — | — |
| A9 | A8 | 70 | VSS_DCDC | — | — | — | — | — | — |
| A10 | A9 | 64 | VCC_DCDC | — | — | — | — | — | — |
| A11 | A10 | 62 | VCC_DCDC | — | — | — | — | — | — |
| A12 | A11 | — | — | P309 | — | IRQ25-DS | CTS9_B/ET1_GTX_CLK/ RGMII1_TXC | GTCPPO8 | VCOUT |
| A13 | B13 | 49 | — | P906 | — | IRQ9 | CTS6_A/USB_ID/ET1_RXD0/ RGMII1_RXD0/RMII1_RXD0/ CAT1_ERXD0 | GTIOC13B/ULPTO1 | AD0FLAG1 |
| A14 | A14 | 50 | — | P905 | — | IRQ8 | RXD3_B/SCL3_B/MISO3_B/ ET1_RX_CLK/RGMII1_RXC/ RMII1_REF50CK/CAT1_RX_CLK | GTCPPO13 | AD1FLAG1 |
| A15 | B14 | 48 | — | P907 | — | IRQ10 | SCK6_A/DE6/USB_EXICEN/ ET1_RXD1/RGMII1_RXD1/ RMII1_RXD1/CAT1_ERXD1 | GTIOC13A/ULPTEE1 | ADSYNC |
| A16 | — | — | — | P904 | — | IRQ2 | ET1_RXD4 | GTIOC11B | — |
| A17 | — | — | — | P207 | — | IRQ25 | ET1_RXD5 | GTCPPO3 | — |
| B1 | E3 | 104 | — | P813 | SDCS | IRQ15 | SCK7_A/DE7 | GTETRGA/GTIOC7B | — |
| B2 | B2 | 89 | — | PA12 | D9/DQ9 | IRQ11 | RXD9_C/SCL9_C/MISO9_C | GTIW/GTIOC6B | — |
| B3 | A3 | 85 | — | P114 | D5/DQ5 | IRQ30-DS | CTS_RTS0_A/SS0_A/DE0/ SSLA0_B/SD0DAT6_B | GTETRGC/GTIOC2B | ADSYNC |
| B4 | A2 | 88 | — | PA11 | D8/DQ8 | IRQ10 | SCK9_C/DE9 | GTIV/GTIOC6A | — |
| B5 | A5 | 82 | — | P300 | D2/DQ2 | IRQ4 | SCK0_A/DE0/SSLA3_B/ SD0DAT3_B | GTIOC3A/ULPTEVI0-DS | — |
| B6 | — | — | — | P303 | — | IRQ29-DS | SCK6_B/DE6 | GTIOC7B | — |
| B7 | B6 | 67 | VLO | — | — | — | — | — | — |
| B8 | B7 | 65 | VLO | — | — | — | — | — | — |
| B9 | B8 | 69 | VSS_DCDC | — | — | — | — | — | — |
| B10 | B9 | 63 | VCC_DCDC | — | — | — | — | — | — |
| B11 | B10 | 61 | VCC_DCDC | — | — | — | — | — | — |
| B12 | B11 | 52 | — | P311 | — | IRQ23-DS | SCK3_B/DE3/CRX0/ET1_TX_CLK/ CAT1_TX_CLK | GTADSM1/GTCPPO6/ AGTOB1 | — |
| B13 | A13 | 47 | — | P908 | — | IRQ11 | TXD6_A/SDA6_A/MOSI6_A/CRX1/ USB_OVRCURB/ET1_RXD2/ RGMII1_RXD2/CAT1_ERXD2 | GTIOC12B/ULPTEVI1 | ADST1 |
| B14 | B15 | 46 | — | P909 | — | IRQ21-DS | RXD6_A/SCL6_A/MISO6_A/CTX1/ USB_OVRCURA/ET1_RXD3/ RGMII1_RXD3/CAT1_ERXD3 | GTIOC12A/ULPTOA1 | ADST0 |
| B15 | A15 | 45 | CLKOUT | P206 | CS7 | IRQ0-DS | USB_VBUSEN/SD0DAT7_C/ ET1_RX_DV/RGMII1_RX_CTL/ RMII1_CRS_DV/CAT1_RX_DV | GTIU/GTCPPO0/ULPTOB1 | — |
| B16 | — | — | — | PD01 | — | IRQ22 | SCK8_C/DE8/SD0DAT2_C/ ET1_RXD6 | GTCPPO2 | — |
| B17 | — | — | — | PD02 | — | IRQ21 | TXD8_C/SDA8_C/MOSI8_C/ SD0DAT1_C/ET1_RXD7 | GTCPPO1 | — |
| C1 | F1 | 106 | — | PA06 | CS1/CKE | IRQ17 | CTS2_C/SD0DAT1_A | GTETRGC/GTIOC7B | — |
| C2 | D2 | 95 | — | P613 | D15/ DQ15 | IRQ19 | CTS0_C | GTETRGA/GTIOC9B/ AGTO1 | — |

**Table 1.15 Pin list for the Standard product (2 of 8)**

| BGA289 | BGA224 | HLQFP176 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/ OSPI/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|---|---|
| C3 | C3 | 90 | — | PA13 | D10/ DQ10 | IRQ12 | CTS_RTS9_C/SS9_C/DE9 | GTOVUP/GTIOC10A | — |
| C4 | C5 | 81 | — | P301 | D1/DQ1 | IRQ6 | TXD6_B/SDA6_B/MOSI6_B/ SD0DAT2_B | GTOULO/GTIOC4B/ AGTIO0/ULPTEE0-DS | — |
| C5 | D5 | 78 | — | P200 | — | NMI | — | — | — |
| C6 | C7 | 72 | TMS/SWDIO | P210 | — | IRQ24 | CTS_RTS9_B/SS9_B/DE9 | GTOULO/GTIOC0B | — |
| C7 | D6 | 74 | TDI | P208 | — | IRQ3 | RXD9_B/SCL9_B/MISO9_B/CRX1 | GTOVLO/GTIOC1B | VCOUT |
| C8 | — | — | — | P110 | — | IRQ20 | SD0DAT4_C | GTIOC9B | — |
| C9 | D10 | 54 | TCLK | P308 | — | IRQ26-DS | CTS3_B/SD0CLK_B/ET1_TX_ER/ ETHPHYCLK | GTIU/GTCPPO9/ULPTOB1 | — |
| C10 | C11 | 58 | TDATA2 | P305 | — | IRQ8 | SD0WP/ET1_TXD2/RGMII1_TXD2/ CAT1_ETXD2 | GTOVUP/GTCPPO12/ ULPTEE1 | — |
| C11 | C12 | 55 | TDATA0 | P307 | — | IRQ27-DS | CTS_RTS6_A/SS6_A/DE6/ SD0CMD_B/ET1_TXD0/ RGMII1_TXD0/RMII1_TXD0/ CAT1_ETXD0 | GTIV/GTCPPO10/ULPTOA1 | — |
| C12 | — | — | — | P911 | — | IRQ6 | ET1_TXD5 | GTIOC3B | — |
| C13 | A12 | 51 | — | P312 | — | IRQ22-DS | CTS_RTS3_B/SS3_B/DE3/CTX0/ ET1_RX_ER/RMII1_RX_ER/ CAT1_RX_ER | GTADSM0/GTCPPO5/ AGTOA1 | — |
| C14 | — | — | — | PD04 | — | IRQ20 | CTS_RTS8_C/SS8_C/DE8/ SD0CMD_C/ET0_RXD5 | GTIOC3A | — |
| C15 | — | — | — | PD03 | — | IRQ21 | RXD8_C/SCL8_C/MISO8_C/ SD0DAT0_C/ET0_RXD4 | GTIOC3B | — |
| C16 | — | — | — | PD05 | — | IRQ19 | CTS8_C/SD0CLK_C/ET0_RXD6 | GTIOC2B | — |
| C17 | — | — | — | PD06 | — | IRQ18 | SD0WP/ET0_RXD7 | GTIOC2A | — |
| D1 | G1 | 108 | — | PA04 | A1/DQM3 | IRQ19 | SCK2_C/DE2/SD0DAT3_A | GTIU/GTIOC4B | ADST0 |
| D2 | C2 | 93 | CACREF/CLKOUT | P611 | D13/ DQ13 | IRQ17 | SCK0_C/DE0/MOSIA_B | GTOULO/GTIOC4B | — |
| D3 | B1 | 92 | — | P610 | D12/ DQ12 | IRQ16 | RXD0_C/SCL0_C/MISO0_C/ RSPCKA_B/CRX1 | GTOUUP/GTIOC4A/ ULPTOB1-DS | — |
| D4 | D4 | 91 | — | PA14 | D11/ DQ11 | IRQ13 | TXD9_C/SDA9_C/MOSI9_C | GTOVLO/GTIOC10B | — |
| D5 | C6 | 76 | RES | — | — | — | — | — | — |
| D6 | C8 | 71 | TCK/SWCLK | P211 | — | IRQ23 | SCK9_B/DE9 | GTOUUP/GTIOC0A | — |
| D7 | — | — | — | P109 | — | IRQ23 | SD0DAT5_C | GTIOC10A | — |
| D8 | — | — | — | P108 | — | IRQ24 | SD0DAT6_C | GTIOC10B | — |
| D9 | — | — | — | P903 | — | IRQ1 | — | GTIOC11A | — |
| D10 | C9 | 60 | TDATA3 | P304 | — | IRQ9 | SD0DAT0_B/ET1_TXD3/ RGMII1_TXD3/CAT1_ETXD3 | GTOVLO/GTIOC7A/ ULPTO1 | — |
| D11 | C10 | 57 | TDATA1 | P306 | — | IRQ28-DS | SD0CD/ET1_TXD1/RGMII1_TXD1/ RMII1_TXD1/CAT1_ETXD1 | GTIW/GTCPPO11/ ULPTEVI1 | — |
| D12 | — | — | — | P912 | — | IRQ5 | ET1_TXD6 | GTIOC3A | — |
| D13 | D12 | 42 | — | PB04 | — | IRQ9 | SCK5_C/DE5/ET0_TXD3/ RGMII0_TXD3/CAT0_ETXD3 | GTCPPO3 | AD0FLAG1 |
| D14 | — | — | — | PB07 | — | IRQ1 | ET0_TXD5 | GTIOC9B | — |
| D15 | — | — | — | PB05 | — | IRQ15 | CTS5_C/ET0_TXD7 | GTCPPO4 | — |
| D16 | C13 | 41 | — | PB03 | — | IRQ13 | TXD5_C/SDA5_C/MOSI5_C/ ET0_TXD2/RGMII0_TXD2/ CAT0_ETXD2 | GTCPPO1 | ADSYNC |
| D17 | C15 | 43 | — | PB01 | ALE | IRQ12 | CTS_RTS1_B/SS1_B/DE1/ ET0_TX_CLK/CAT0_TX_CLK | GTCPPO2 | AD1FLAG1 |
| E1 | E1 | 103 | — | PA15 | EBCLK/ SDCLK | IRQ14 | CTS9_C | GTIOC7A | — |
| E2 | D1 | 97 | — | P615 | WR2/BC 2/DQM2 | IRQ7 | TXD7_A/SDA7_A/MOSI7_A | GTETRGC/GTCPPO10 | — |
| E3 | D3 | 96 | — | P614 | WR/WR0 /DQM0 | IRQ20 | RXD7_A/SCL7_A/MISO7_A | GTETRGB/GTCPPO9/ AGTO0 | — |
| E4 | C1 | 94 | — | P612 | D14/ DQ14 | IRQ18 | CTS_RTS0_C/SS0_C/DE0/ SSLA0_B | GTIOC9A | — |
| E5 | — | — | — | P914 | — | IRQ9 | CTS_RTS6_B/SS6_B/DE6 | GTIOC5B | — |
| E6 | D7 | 77 | MD | P201 | — | IRQ4 | — | — | — |

### Table 1.15 Pin list for the Standard product (3 of 8)

| BGA289 | BGA224 | HLQFP176 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/ OSPI/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|---|---|
| E7 | D8 | 73 | TDO/SWO/CLKOUT | P209 | — | IRQ25 | TXD9_B/SDA9_B/MOSI9_B/CTX1 | GTOVUP/GTIOC1A | — |
| E8 | — | — | — | P111 | — | IRQ19 | SD0DAT3_C | GTIOC9A | — |
| E9 | D9 | — | — | P902 | ALE | IRQ0 | ETHPHYCLK | GTCPPO13 | — |
| E10 | B12 | 53 | — | P310 | — | IRQ24-DS | TXD3_B/SDA3_B/MOSI3_B/ ET1_TX_EN/RGMII1_TX_CTL/ RMII1_TX_EN/CAT1_TX_EN | GTCPPO7/AGTEE1 | — |
| E11 | — | — | — | P910 | — | IRQ7 | ET1_TXD4 | GTCPPO12 | — |
| E12 | — | — | CLKOUT | P913 | — | IRQ3 | ET1_TXD7 | GTCPPO11 | — |
| E13 | D11 | 40 | — | PB02 | — | IRQ11 | RXD5_C/SCL5_C/MISO5_C/ ET0_TXD1/RGMII0_TXD1/ RMII0_TXD1/CAT0_ETXD1 | GTCPPO0 | ADST1 |
| E14 | — | — | — | PB06 | — | IRQ0 | CTS_RTS5_C/SS5_C/DE5/ ET0_TXD6 | GTIOC9A | — |
| E15 | — | — | — | PD07 | — | IRQ17 | SD0CD/ET0_TXD4 | GTCPPO0 | — |
| E16 | C14 | 39 | — | PB00 | — | IRQ10 | SCK1_B/DE1/ET0_TXD0/ RGMII0_TXD0/RMII0_TXD0/ CAT0_ETXD0/DSM1CLK2 | GTCPPO4 | ADST0 |
| E17 | D15 | — | — | P706 | — | IRQ7 | RXD1_B/SCL1_B/MISO1_B/ ET0_GTX_CLK/RGMII0_TXC/ ETHPHYCLK/DSM1CLK0 | GTCPPO2/AGTIO0 | — |
| F1 | G2 | 110 | — | PA02 | A3 | IRQ31 | RXD2_C/SCL2_C/MISO2_C/ SD0DAT5_A | GTIW/GTCPPO9 | ADSYNC |
| F2 | F2 | 102 | — | PA10 | CS2/RAS | IRQ4 | SCK5_B/DE5 | GTCPPO13 | — |
| F3 | E2 | 100 | — | PA08 | CS0/WE | IRQ6 | RXD5_B/SCL5_B/MISO5_B | GTETRGD/GTCPPO11 | — |
| F4 | E4 | 101 | — | PA09 | CS3/CAS | IRQ5 | TXD5_B/SDA5_B/MOSI5_B | GTCPPO12 | — |
| F5 | H5 | 125 | — | PC14 | D16/ DQ16 | IRQ0 | TXD6_C/SDA6_C/MOSI6_C/ ET0_WOL/DSM1DAT1 | GTADSM1/GTCPPO9 | — |
| F6 | E5 | 79 | VCC_08 | — | — | — | — | — | — |
| F7 | E6 | — | VSS_08 | — | — | — | — | — | — |
| F8 | G10 | — | VSS3 | — | — | — | — | — | — |
| F9 | G9 | 75 | VCL3 | — | — | — | — | — | — |
| F10 | E9 | — | VSS_07 | — | — | — | — | — | — |
| F11 | E10 | 56 | VCC_07 | — | — | — | — | — | — |
| F12 | F11 | 31 | — | P700 | — | IRQ16-DS | RXD2_B/SCL2_B/MISO2_B/ MISOA_C/SD1WP/ET0_RXD2/ RGMII0_RXD2/CAT0_ERXD2/ DSM0CLK0 | GTIOC5A | — |
| F13 | E12 | 33 | — | P702 | — | IRQ18-DS | CTS2_B/RSPCKA_C/SD1DAT5_B/ ET0_RXD0/RGMII0_RXD0/ RMII0_RXD0/CAT0_ERXD0/ DSM0CLK2 | GTIOC6A/ULPTO0 | — |
| F14 | E14 | 30 | — | P406 | — | IRQ31 | TXD2_B/SDA2_B/MOSI2_B/ SSLA3_C/SD1CD/ET0_RXD3/ RGMII0_RXD3/CAT0_ERXD3/ DSM0DAT2 | GTIOC1B | — |
| F15 | E15 | 32 | — | P701 | — | IRQ17-DS | CTS_RTS2_B/SS2_B/DE2/ MOSIA_C/SD1DAT4_B/ET0_RXD1/ RGMII0_RXD1/RMII0_RXD1/ CAT0_ERXD1/DSM0CLK1 | GTIOC5B/ULPTO1 | — |
| F16 | D14 | 37 | — | P707 | — | IRQ8 | TXD1_B/SDA1_B/MOSI1_B/ ET0_TX_ER/ETHPHYCLK/ DSM1CLK1 | GTCPPO3 | — |
| F17 | D13 | 36 | — | P705 | — | IRQ19 | CTS1_B/SSLA2_C/CRX0/ ET0_TX_EN/RGMII0_TX_CTL/ RMII0_TX_EN/CAT0_TX_EN/ DSM1DAT2 | GTADSM1/GTCPPO1/ AGTIO0 | — |
| G1 | H1 | 112 | — | PA00 | A5 | IRQ22 | CTS_RTS5_B/SS5_B/DE5/ SD0DAT7_A | GTOVLO/GTCPPO7 | AD1FLAG1 |
| G2 | G4 | 109 | — | PA03 | A2 | IRQ20 | TXD2_C/SDA2_C/MOSI2_C/ SD0DAT4_A | GTIV/GTCPPO10 | ADST1 |
| G3 | F3 | 107 | — | PA05 | A0/BC0/ DQM1 | IRQ18 | CTS_RTS2_C/SS2_C/DE2/ SD0DAT2_A | GTETRGD/GTIOC4A | — |
| G4 | F4 | 105 | — | PA07 | RD | IRQ16 | CTS7_A/SD0DAT0_A | GTETRGB/GTIOC7A | VCOUT |
| G5 | J5 | 127 | — | PC12 | D18/ DQ18 | IRQ2 | SCK6_C/DE6/ET0_MDIO/ CAT0_MDIO/DSM1CLK0 | GTCPPO11 | — |
| G6 | F5 | 98 | VCC_09 | — | — | — | — | — | — |

### Table 1.15 Pin list for the Standard product (4 of 8)

| BGA289 | BGA224 | HLQFP176 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/ OSPI/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|---|---|
| G7 | F6 | — | VSS_09 | — | — | — | — | — | — |
| G8 | F10 | — | VSS4 | — | — | — | — | — | — |
| G9 | F9 | — | VCL4 | — | — | — | — | — | — |
| G10 | — | — | VSS_06 | — | — | — | — | — | — |
| G11 | — | — | VCC_06 | — | — | — | — | — | — |
| G12 | E11 | 29 | — | P405 | — | IRQ30 | SCK2_B/DE2/SD1DAT3_B/ ET0_RX_DV/RGMII0_RX_CTL/ RMII0_CRS_DV/CAT0_RX_DV/ DSM0DAT1 | GTIOC1A/AGTIO1 | — |
| G13 | E13 | 35 | — | P704 | — | IRQ26 | SSLA1_C/CTX0/SD1DAT7_B/ ET0_RX_ER/RMII0_RX_ER/ CAT0_RX_ER/DSM1DAT1 | GTADSM0/GTCPPO0/ AGTO0 | — |
| G14 | F12 | 34 | — | P703 | — | IRQ19-DS | SSLA0_C/SD1DAT6_B/ ET0_RX_CLK/RGMII0_RXC/ RMII0_REF50CK/CAT0_RX_CLK/ DSM1DAT0 | GTIOC6B/AGTO1 | VCOUT |
| G15 | F13 | 28 | — | PB14 | — | IRQ10 | CATSYNC0 | — | — |
| G16 | F14 | 26 | — | PB12 | — | IRQ6 | CATLINKACT0 | — | — |
| G17 | F15 | 25 | — | PB11 | — | IRQ5 | CATLEDERR | — | — |
| H1 | H2 | 114 | — | P504 | A7 | IRQ7 | SD0WP | GTOULO/GTCPPO1 | — |
| H2 | H3 | 113 | — | P503 | A6 | IRQ6 | SD0CD | GTOUUP/GTCPPO6 | — |
| H3 | H4 | 115 | — | P505 | A8 | IRQ8 | SD0CLK_A | GTOWUP/GTCPPO2 | — |
| H4 | G3 | 111 | — | PA01 | A4 | IRQ21 | CTS5_B/SD0DAT6_A | GTOVUP/GTCPPO8 | AD0FLAG1 |
| H5 | M1 | 128 | — | PC11 | D19/ DQ19 | IRQ3 | CTS_RTS6_C/SS6_C/DE6/ ET0_MDC/CAT0_MDC/DSM1CLK1 | GTCPPO12 | — |
| H6 | G5 | 118 | VCC_10 | — | — | — | — | — | — |
| H7 | G6 | — | VSS_10 | — | — | — | — | — | — |
| H8 | G7 | — | VSS7 | — | — | — | — | — | — |
| H9 | E8 | — | VCL5 | — | — | — | — | — | — |
| H10 | E7 | — | VSS5 | — | — | — | — | — | — |
| H11 | — | — | VCC_04 | — | — | — | — | — | — |
| H12 | — | — | VSS_04 | — | — | — | — | — | — |
| H13 | H11 | 13 | — | P403 | — | IRQ14-DS | CTS_RTS1_A/SS1_A/DE1/ SD1DAT1_B/ET1_WOL/ CATI2CCLK | GTIOC3A/RTCIC1 | AD0FLAG1 |
| H14 | G12 | — | VCC_05 | — | — | — | — | — | — |
| H15 | G13 | 23 | — | PB09 | — | IRQ13 | CATIRQ | — | — |
| H16 | G14 | 24 | — | PB10 | — | IRQ4 | CATLEDSTER | — | — |
| H17 | G15 | 27 | — | PB13 | — | IRQ7 | CATLINKACT1 | — | — |
| J1 | J1 | 116 | — | P506 | A9 | IRQ9 | SD0CMD_A | GTOWLO/GTCPPO3 | — |
| J2 | J3 | 117 | — | P507 | A10 | IRQ10 | CTS_RTS7_A/SS7_A/DE7/ ET_TAS_STA0 | GTADSM0/GTIOC0A | — |
| J3 | J4 | 119 | — | P508 | A11 | IRQ1 | CTS5_A/ET_TAS_STA1 | GTADSM1/GTIOC0B | — |
| J4 | K4 | 120 | — | P509 | A12 | IRQ2 | CTS_RTS5_A/SS5_A/DE5/ ET_TAS_STA2 | GTIOC1A/ULPTEVI1 | — |
| J5 | L1 | 126 | — | PC13 | D17/ DQ17 | IRQ1 | RXD6_C/SCL6_C/MISO6_C/ ET0_INT/DSM1DAT2 | GTCPPO10 | — |
| J6 | — | — | VCC2_11 | — | — | — | — | — | — |
| J7 | — | — | VSS_11 | — | — | — | — | — | — |
| J8 | G8 | — | VCL7 | — | — | — | — | — | — |
| J9 | F8 | — | VCL6 | — | — | — | — | — | — |
| J10 | F7 | — | VSS6 | — | — | — | — | — | — |
| J11 | H9 | 59 | VCL2 | — | — | — | — | — | — |
| J12 | H10 | — | VSS2 | — | — | — | — | — | — |
| J13 | G11 | 14 | — | P404 | — | IRQ15-DS | CTS1_A/SD1DAT2_B/ET0_WOL/ CATI2CDATA/DSM0DAT0 | GTIOC3B/RTCIC2 | AD1FLAG1 |
| J14 | H12 | 22 | — | PB08 | — | IRQ12 | CATLEDRUN | — | — |
| J15 | H13 | — | VSS_05 | — | — | — | — | — | — |

### Table 1.15 Pin list for the Standard product (5 of 8)

| BGA289 | BGA224 | HLQFP176 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/ OSPI/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|---|---|
| J16 | H14 | — | VSS_03 | — | — | — | — | — | — |
| J17 | H15 | — | VSS_02 | — | — | — | — | — | — |
| K1 | K1 | 124 | — | PC15 | A16 | IRQ30 | CTS6_C/CRX1 | GTADSM0 | — |
| K2 | K2 | 122 | CACREF | P608 | A14 | IRQ22 | TXD5_A/SDA5_A/MOSI5_A | GTOWUP/GTCPPO4 | — |
| K3 | J2 | 121 | — | P510 | A13 | IRQ3 | RXD5_A/SCL5_A/MISO5_A/ ET_TAS_STA3 | GTIOC1B/ULPTEVI0 | — |
| K4 | K3 | 123 | — | PD00 | A15 | IRQ23 | SCK5_A/DE5/CTX1 | GTOWLO/GTCPPO5 | — |
| K5 | — | — | — | PC07 | D23/ DQ23 | IRQ21 | OM_1_RESET/DSM0DAT0 | GTCPPO0 | — |
| K6 | R4 | — | VSS_12 | — | — | — | — | — | — |
| K7 | J7 | — | VSS9 | — | — | — | — | — | — |
| K8 | J8 | 129 | VCL9 | — | — | — | — | — | — |
| K9 | H8 | 99 | VCL8 | — | — | — | — | — | — |
| K10 | H7 | — | VSS8 | — | — | — | — | — | — |
| K11 | J9 | 44 | VCL1 | — | — | — | — | — | — |
| K12 | J10 | — | VSS1 | — | — | — | — | — | — |
| K13 | K11 | 6 | — | P410 | A19 | IRQ5 | SCK3_A/DE3/SCL0_A[*1]/ USB_OVRCURB-DS/ GPTP_MATCH0/CATI2CCLK | GTOVLO/GTIOC9B/ AGTOB1 | ADST0 |
| K14 | J12 | 38 | VCC_03 | — | — | — | — | — | — |
| K15 | J13 | 21 | VCC_02 | — | — | — | — | — | — |
| K16 | J14 | 19 | XTAL | P213 | — | IRQ2 | TXD1_C/SDA1_C/MOSI1_C | GTETRGC/GTIOC0A/ ULPTEE0 | ADTRG1 |
| K17 | J15 | 20 | EXTAL | P212 | — | IRQ3 | RXD1_C/SCL1_C/MISO1_C | GTETRGD/GTIOC0B/ AGTEE1 | — |
| L1 | — | — | — | PC03 | D27/ DQ27 | IRQ25 | TXD7_C/SDA7_C/MOSI7_C/ OM_1_SIO4/DSM0CLK1 | GTCPPO4 | — |
| L2 | — | — | — | PC02 | D28/ DQ28 | IRQ26 | SCK7_C/DE7/OM_1_SIO3/ DSM0CLK2 | GTCPPO5 | — |
| L3 | — | — | — | PC04 | D26/ DQ26 | IRQ24 | RXD7_C/SCL7_C/MISO7_C/ OM_1_SIO2/DSM0CLK0 | GTCPPO3 | — |
| L4 | — | — | — | PC09 | D21/ DQ21 | IRQ5 | OM_1_RSTO1 | — | — |
| L5 | — | — | — | PC05 | D25/ DQ25 | IRQ23 | OM_1_CS1/DSM0DAT2 | GTCPPO2 | — |
| L6 | N1 | 130 | VCC2_12 | — | — | — | — | — | — |
| L7 | K6 | — | VSS_14 | — | — | — | — | — | — |
| L8 | H6 | — | VSS_15 | — | — | — | — | — | — |
| L9 | K7 | — | VSS10 | — | — | — | — | — | — |
| L10 | K8 | 151 | VCL10 | — | — | — | — | — | — |
| L11 | K9 | 18 | VCL0 | — | — | — | — | — | — |
| L12 | K10 | — | VSS0 | — | — | — | — | — | — |
| L13 | L11 | 5 | — | P414 | A23 | IRQ9 | RXD4_B/SCL4_B/MISO4_B/ SSLB0_B/CRX1/ET1_MDIO/ CATLATCH1 | GTIOC0B | — |
| L14 | J11 | 12 | CACREF | P402 | — | IRQ4-DS | SCK1_A/DE1/CRX0/SD1DAT0_B/ ET0_LINKSTA/CAT0_LINKSTA | RTCIC0 | — |
| L15 | K13 | 1 | VCC_01 | — | — | — | — | — | — |
| L16 | K14 | 17 | XCOUT | P214 | — | IRQ21 | — | — | — |
| L17 | K15 | 16 | XCIN/EXCIN | P215 | — | IRQ20 | — | — | — |
| M1 | — | — | — | PC00 | D30/ DQ30 | IRQ28 | CTS_RTS7_C/SS7_C/DE7/ OM_1_SIO5 | GTCPPO7 | — |
| M2 | — | — | — | P607 | D31/ DQ31 | IRQ23 | OM_1_DQS | — | — |
| M3 | — | — | — | PC01 | D29/ DQ29 | IRQ27 | CTS7_C/OM_1_SIO0 | GTCPPO6 | — |
| M4 | — | — | — | PC08 | D22/ DQ22 | IRQ29 | OM_1_CS0/DSM1DAT0 | GTCPPO8 | — |
| M5 | — | — | — | PC10 | D20/ DQ20 | IRQ4 | OM_1_WP1/DSM1CLK2 | GTCPPO13 | — |

### Table 1.15 Pin list for the Standard product (6 of 8)

| BGA289 | BGA224 | HLQFP176 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/ OSPI/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|---|---|
| M6 | L6 | 137 | — | P104 | — | IRQ1 | CTS9_A/SSLB1_A/OM_0_CS1/ GPTP_MATCH0/CATLATCH0/ DSM0DAT1 | GTETRGB/GTIOC1B | AD0FLAG1 |
| M7 | K5 | 141 | VCC2_14 | — | — | — | — | — | — |
| M8 | J6 | 147 | VCC2_15 | — | — | — | — | — | — |
| M9 | — | — | — | P810 | — | IRQ21 | SCK7_B/DE7/SD1DAT2_A | GTIOC10A/ULPTOA0 | — |
| M10 | L8 | — | VSS11 | — | — | — | — | — | — |
| M11 | L9 | 171 | VCL11 | — | — | — | — | — | — |
| M12 | — | — | — | P412 | A21 | IRQ20-DS | CTS3_A/USB_EXICEN/ GPTP_PTPOUT0 | GTOULO/GTCPPO8/ AGTEE1 | — |
| M13 | M12 | — | — | P710 | CS5 | IRQ17 | CTS4_B/SSLB3_B/ET0_LINKSTA/ CAT0_LINKSTA | GTIOC11B | — |
| M14 | M11 | — | CACREF | P411 | A20 | IRQ4 | CTS_RTS3_A/SS3_A/DE3/USB_ID/ GPTP_PTPOUT1 | GTOVUP/GTIOC9A/ AGTOA1 | — |
| M15 | L12 | 8 | — | P408 | A17 | IRQ7 | RXD3_A/SCL3_A/MISO3_A/ SCL0_B[*1]/USB_VBUSEN/ GPTP_PTPOUT2/CATRESETOUT | GTOWLO/GTIOC10A/ ULPTOB0 | ADSYNC |
| M16 | L14 | 15 | VBATT | — | — | — | — | — | — |
| M17 | L15 | — | VSS_01 | — | — | — | — | — | — |
| N1 | — | — | — | P605 | — | IRQ25 | CTS0_B/OM_1_SIO1 | GTIOC8A | — |
| N2 | L2 | — | — | P604 | — | IRQ26 | CTS_RTS0_B/SS0_B/DE0/ OM_1_SIO7 | GTIOC8B | — |
| N3 | — | — | — | P606 | WR3/BC 3 | IRQ24 | OM_1_SIO6 | — | — |
| N4 | — | — | — | PC06 | D24/ DQ24 | IRQ22 | OM_1_ECSINT1/DSM0DAT1 | GTCPPO1 | — |
| N5 | L4 | 134 | — | P107 | — | IRQ31 | CTS4_A/OM_0_CS0/ET1_INT | GTOWUP/GTIOC8A/ AGTOA0 | ADST0 |
| N6 | L5 | 135 | — | P106 | — | IRQ16 | CTS8_B/SSLB3_A/OM_0_RESET/ ET1_LINKSTA/CAT1_LINKSTA | GTOWLO/GTIOC8B/ AGTOB0/ULPTEE1-DS | ADST1 |
| N7 | L7 | 136 | — | P105 | — | IRQ0 | CTS_RTS8_B/SS8_B/DE8/ SSLB2_A/OM_0_ECSINT1/ GPTP_CAPTURE0/CATSYNC1/ DSM0DAT0 | GTIOC1A/ULPTO1-DS | ADSYNC |
| N8 | — | — | — | P811 | — | IRQ22 | CTS7_B/USB_ID/SD1DAT3_A | GTIOC10B/ULPTOB0 | — |
| N9 | — | — | — | P013 | — | IRQ14 | — | — | AN013 |
| N10 | — | — | — | P011 | — | IRQ16 | — | — | AN011 |
| N11 | — | — | — | P807 | — | IRQ11 | — | GTIOC13A | — |
| N12 | M10 | 3 | CACREF | P708 | WR1/BC 1 | IRQ11 | SCK4_B/DE4/SDA2_A[*1]/MOSIB_B/ ET0_MDC/CAT0_MDC | GTCPPO6 | — |
| N13 | N12 | — | — | P712 | — | IRQ2 | CTS1_C/SSLB1_B/ GPTP_CAPTURE1/CATLATCH1 | GTIOC2B/AGTOB0 | — |
| N14 | N13 | — | — | P714 | — | IRQ13 | TXD4_C/SDA4_C/MOSI4_C/ GPTP_PPS1/CATSYNC1 | GTIOC12B | — |
| N15 | M14 | — | — | P711 | — | IRQ3 | CTS_RTS1_C/SS1_C/DE1/ SSLB2_B/GPTP_PPS0/ CATRESETOUT | GTIOC11A/AGTEE0 | — |
| N16 | N14 | — | — | P713 | — | IRQ14 | CTS4_C/GPTP_MATCH1/ CATLATCH0 | GTIOC2A/AGTOA0 | — |
| N17 | M15 | 11 | — | P401 | — | IRQ5-DS | RXD1_A/SCL1_A/MISO1_A/ I3C_SDA0/CTX0/SD1CMD_B | GTETRGA/GTIOC6B | — |
| P1 | L3 | — | — | P603 | — | IRQ27 | TXD0_B/SDA0_B/MOSI0_B/ OM_1_SCLK | GTIOC7A/ULPTO0 | — |
| P2 | M2 | — | — | P602 | — | IRQ28 | RXD0_B/SCL0_B/MISO0_B/ OM_1_SCLKN | GTIOC7B/ULPTEE0 | — |
| P3 | M3 | 132 | CACREF | P600 | — | IRQ30 | OM_0_RSTO1/ET1_WOL | GTIOC6B/ULPTEVI1-DS | — |
| P4 | M4 | 131 | — | P601 | — | IRQ29 | SCK0_B/DE0/OM_0_WP1 | GTIOC6A/ULPTEVI0/ RTCOUT | — |
| P5 | M5 | 139 | — | P102 | — | IRQ17 | TXD9_A/SDA9_A/MOSI9_A/ RSPCKB_A/CRX0/OM_0_SIO4/ DSM0CLK0 | GTOWLO/GTIOC2B/AGTO0 | ADTRG0 |
| P6 | M6 | 144 | — | P801 | — | IRQ12 | TXD2_A/SDA2_A/MOSI2_A/ OM_0_DQS/GPTP_PPS1/ CATRESETOUT/DSM1DAT1 | GTIV/GTIOC11B/AGTOB0 | — |

### Table 1.15 Pin list for the Standard product (7 of 8)

| BGA289 | BGA224 | HLQFP176 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/ OSPI/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|---|---|
| P7 | M7 | 146 | — | P803 | — | IRQ19 | SCK2_A/DE2/OM_0_SIO1/ DSM1CLK0 | GTETRGC/GTIOC12B | — |
| P8 | — | — | — | P812 | — | IRQ23 | CTS_RTS7_B/SS7_B/DE7/ USB_EXICEN/SD1DAT4_A | GTIOC11A | AN022 |
| P9 | — | — | — | P012 | — | IRQ15 | — | — | AN012 |
| P10 | — | — | — | P010 | — | IRQ14 | — | — | AN010 |
| P11 | M8 | 160 | — | P009 | — | IRQ13-DS | — | — | AN009/IVREF1 |
| P12 | — | — | — | P805 | — | IRQ30 | TXD8_A/SDA8_A/MOSI8_A/ ET1_MDIO | — | AN017/IVCMP0 |
| P13 | P13 | 170 | — | P512 | — | IRQ14 | CTS8_A/SCL1_A[*1]/CTX1/ET1_INT | GTIOC0A | — |
| P14 | — | — | — | P413 | A22 | IRQ18 | ET_TAS_STA3 | GTOUUP/GTCPPO7/ ULPTEE1 | — |
| P15 | — | — | — | P515 | — | IRQ12 | CTS_RTS4_C/SS4_C/DE4/ SCL2_B[*1]/ET_TAS_STA0 | GTIOC13A | — |
| P16 | M13 | 2 | — | P709 | CS4 | IRQ10 | CTS_RTS4_B/SS4_B/DE4/ SCL2_A[*1]/MISOB_B/ET0_MDIO/ CAT0_MDIO | GTCPPO5 | — |
| P17 | N15 | 10 | — | P400 | — | IRQ0 | TXD1_A/SDA1_A/MOSI1_A/ I3C_SCL0/SD1CLK_B | GTIOC6A/AGTIO1 | ADTRG1 |
| R1 | P4 | 133 | VCC2_13 | — | — | — | — | — | — |
| R2 | N2 | — | — | P315 | — | IRQ29 | SCK3_C/DE3/SSLA3_A | — | — |
| R3 | — | — | — | P900 | — | IRQ30 | CTS3_C | GTADSM0 | — |
| R4 | N4 | 138 | — | P103 | — | IRQ16 | CTS_RTS9_A/SS9_A/DE9/ SSLB0_A/CTX0/OM_0_SIO2/ GPTP_PPS0/CATLATCH1/ DSM0DAT2 | GTOWUP/GTIOC2A | AD1FLAG1 |
| R5 | N5 | 140 | — | P101 | — | IRQ1 | RXD9_A/SCL9_A/MISO9_A/ MOSIB_A/OM_0_SIO3/ GPTP_CAPTURE1/CATI2CCLK/ DSM0CLK1 | GTETRGB/GTIOC8A/ AGTEE0 | — |
| R6 | N6 | 145 | — | P802 | — | IRQ18 | RXD2_A/SCL2_A/MISO2_A/ OM_0_SIO6/DSM1DAT2 | GTIW/GTIOC12A | — |
| R7 | N7 | 148 | — | P804 | — | IRQ14 | CTS_RTS2_A/SS2_A/DE2/ OM_0_SIO7/DSM1CLK1 | GTETRGD/GTIOC13A | — |
| R8 | — | — | — | P501 | — | IRQ25 | TXD8_B/SDA8_B/MOSI8_B/ USB_OVRCURA/SD1DAT6_A | GTIOC12A | AN020 |
| R9 | N8 | 156 | AVCC0 | — | — | — | — | — | — |
| R10 | N9 | 157 | AVSS0 | — | — | — | — | — | — |
| R11 | N10 | 164 | — | P005 | — | IRQ10-DS | — | — | AN005/IVCMP3 |
| R12 | P12 | 166 | — | P003 | — | IRQ29 | — | — | AN003/IVCMP3 |
| R13 | — | — | — | P513 | — | IRQ31 | SCK8_A/DE8/ET0_INT | GTIOC13B | AN016/IVCMP0 |
| R14 | — | — | — | P514 | — | IRQ13 | SCK4_C/DE4/SDA2_B[*1]/ ET_TAS_STA1 | GTIOC13B | — |
| R15 | L13 | 4 | — | P415 | WAIT | IRQ8 | TXD4_B/SDA4_B/MOSI4_B/ RSPCKB_B/CTX1/ET1_MDC/ CATLATCH0 | GTIOC0A | — |
| R16 | L10 | 7 | — | P409 | A18 | IRQ6 | TXD3_A/SDA3_A/MOSI3_A/ SDA0_A[*1]/USB_OVRCURA-DS/ GPTP_CAPTURE0/CATI2CDATA | GTOWUP/ULPTOA0 | ADST1 |
| R17 | K12 | 9 | — | P407 | CS6 | IRQ22 | SCK1_C/DE1/SDA0_B[*1]/ USB_VBUS/GPTP_PTPOUT3/ CATSYNC1 | GTIOC10B/AGTIO0/ RTCOUT | ADTRG0 |
| T1 | P1 | — | CLKOUT | P205 | — | IRQ1-DS | TXD4_A/SDA4_A/MOSI4_A/ SCL1_B[*1]/SSLA1_A/ USB_OVRCURA/SD1CD | GTIV/GTIOC4A/AGTO1 | — |
| T2 | P2 | — | — | P203 | — | IRQ2-DS | RXD4_A/SCL4_A/MISO4_A/ RSPCKA_A/CTX0/USB_VBUSEN/ SD1CLK_A | GTIOC5A/ULPTOA1 | — |
| T3 | P3 | — | — | P313 | — | IRQ27 | TXD3_C/SDA3_C/MOSI3_C/ MISOA_A/USB_ID/SD1DAT0_A | — | — |
| T4 | — | — | — | P901 | — | IRQ31 | CTS_RTS3_C/SS3_C/DE3 | GTADSM1/AGTIO1 | — |
| T5 | P5 | 150 | — | P809 | — | IRQ20 | TXD7_B/SDA7_B/MOSI7_B/ OM_0_SCLKN | — | — |
| T6 | P6 | 143 | — | P800 | — | IRQ11 | CTS2_A/OM_0_SIO5/DSM1DAT0 | GTIU/GTIOC11A/AGTOA0 | — |

### Table 1.15 Pin list for the Standard product (8 of 8)

| BGA289 | BGA224 | HLQFP176 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/ OSPI/SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|---|---|
| T7 | — | — | — | P502 | — | IRQ26 | SCK8_B/DE8/USB_OVRCURB/ SD1DAT7_A | GTIOC12B | AN019 |
| T8 | R7 | 153 | — | P014 | — | IRQ27 | — | — | AN014/DA0/ IVCMP0 |
| T9 | P8 | 154 | VREFL | — | — | — | — | — | — |
| T10 | P9 | 158 | VREFL0 | — | — | — | — | — | — |
| T11 | R11 | 165 | — | P004 | — | IRQ9-DS | — | — | AN004/IVCMP2 |
| T12 | M9 | 162 | — | P007 | — | IRQ28 | — | — | AN007/IVCMP3 |
| T13 | N11 | 168 | — | P001 | — | IRQ7-DS | — | — | AN001/IVCMP3 |
| T14 | — | — | — | P806 | — | IRQ0 | RXD8_A/SCL8_A/MISO8_A/ ET1_MDC | — | AN018 |
| T15 | — | — | — | P715 | — | IRQ12 | RXD4_C/SCL4_C/MISO4_C/ ET_TAS_STA2 | GTIOC12A | — |
| T16 | P14 | 174 | — | P815 | — | IRQ15 | CTX0/USB_DM | GTIOC8A | — |
| T17 | P15 | 173 | VSS_USB | — | — | — | — | — | — |
| U1 | R1 | — | CACREF | P204 | — | IRQ26 | SCK4_A/DE4/SDA1_B*1/SSLA0_A/ USB_OVRCURB/SD1WP | GTIW/GTIOC4B/AGTIO1 | — |
| U2 | R2 | — | — | P202 | — | IRQ3-DS | CTS_RTS4_A/SS4_A/DE4/ MOSIA_A/CRX0/USB_EXICEN/ SD1CMD_A | GTIOC5B/ULPTOB1 | — |
| U3 | R3 | — | — | P314 | — | IRQ28 | RXD3_C/SCL3_C/MISO3_C/ SSLA2_A/SD1DAT1_A | — | ADTRG0 |
| U4 | N3 | — | VSS_13 | — | — | — | — | — | — |
| U5 | R5 | 149 | — | P808 | — | IRQ15 | RXD7_B/SCL7_B/MISO7_B/ OM_0_SCLK/DSM1CLK2 | GTIOC13B | — |
| U6 | R6 | 142 | — | P100 | — | IRQ2 | SCK9_A/DE9/MISOB_A/ OM_0_SIO0/GPTP_MATCH1/ CATI2CDATA/DSM0CLK2 | GTETRGA/GTIOC8B/ AGTIO0 | — |
| U7 | — | — | CACREF | P500 | — | IRQ24 | RXD8_B/SCL8_B/MISO8_B/ USB_VBUSEN/SD1DAT5_A | GTIOC11B | AN021 |
| U8 | P7 | 152 | — | P015 | — | IRQ13 | — | — | AN015/DA1/ IVCMP0 |
| U9 | R8 | 155 | VREFH | — | — | — | — | — | — |
| U10 | R9 | 159 | VREFH0 | — | — | — | — | — | — |
| U11 | R10 | 161 | — | P008 | — | IRQ12-DS | — | — | AN008/IVREF0 |
| U12 | P10 | 163 | — | P006 | — | IRQ11-DS | — | — | AN006/IVCMP2 |
| U13 | R12 | 169 | — | P000 | — | IRQ6-DS | — | — | AN000/IVCMP2 |
| U14 | P11 | 167 | — | P002 | — | IRQ8-DS | — | — | AN002/IVCMP2 |
| U15 | R13 | 172 | — | P511 | — | IRQ15 | CTS_RTS8_A/SS8_A/DE8/ SDA1_A*1/CRX1/ET1_LINKSTA/ CAT1_LINKSTA | GTIOC0B | — |
| U16 | R14 | 175 | — | P814 | — | IRQ16 | CRX0/USB_DP | GTIOC8B | — |
| U17 | R15 | 176 | VCC_USB | — | — | — | — | — | — |

Note: Several pin names have the added suffix of _A, _B, and _C. These suffixes have special conditions for electrical characteristics. See section 60, Electrical Characteristics for detail.

Note 1. There are two types for IIC function, one is a simple IIC by SCI and another is a dedicated IIC. This terminal is for a dedicated IIC.

### Table 1.16 Pin list for the SiP product (1 of 9)

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| A1 | VSS | — | — | — | — | — | — |
| A2 | — | P114 | D5/DQ5 | IRQ30-DS | CTS_RTS0_A/SS0_A/DE0/SSLA0_B/ SD0DAT6_B | GTETRGC/GTIOC2B | ADSYNC |
| A3 | — | P609 | D7/DQ7 | IRQ29 | TXD0_C/SDA0_C/MOSI0_C/MISOA_B/CTX1 | GTIU/GTIOC5B/ULPTOA1-DS | AD1FLAG1 |
| A4 | — | P113 | D4/DQ4 | IRQ28 | RXD0_A/SCL0_A/MISO0_A/SSLA1_B/ SD0DAT5_B | GTETRGB/GTIOC2A/ULPTOA0-DS | ADST1 |
| A5 | — | P301 | D1/DQ1 | IRQ6 | TXD6_B/SDA6_B/MOSI6_B/SD0DAT2_B | GTOULO/GTIOC4B/AGTIO0/ ULPTEE0-DS | — |
| A6 | TDI | P208 | — | IRQ3 | RXD9_B/SCL9_B/MISO9_B/CRX1 | GTOVLO/GTIOC1B | VCOUT |
| A7 | TMS/SWDIO | P210 | — | IRQ24 | CTS_RTS9_B/SS9_B/DE9 | GTOULO/GTIOC0B | — |

**Table 1.16 Pin list for the SiP product (2 of 9)**

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| A8 | VLO | — | — | — | — | — | — |
| A9 | VLO | — | — | — | — | — | — |
| A10 | VSS_DCDC | — | — | — | — | — | — |
| A11 | VCC_DCDC | — | — | — | — | — | — |
| A12 | VCC_DCDC | — | — | — | — | — | — |
| A13 | — | P309 | — | IRQ25-DS | CTS9_B/ET1_GTX_CLK/RGMII1_TXC | GTCPPO8 | VCOUT |
| A14 | — | P906 | — | IRQ9 | CTS6_A/USB_ID/ET1_RXD0/RGMII1_RXD0/ RMII1_RXD0/CAT1_ERXD0 | GTIOC13B/ULPTO1 | AD0FLAG1 |
| A15 | — | P905 | — | IRQ8 | RXD3_B/SCL3_B/MISO3_B/ET1_RX_CLK/ RGMII1_RXC/RMII1_REF50CK/ CAT1_RX_CLK | GTCPPO13 | AD1FLAG1 |
| A16 | — | P907 | — | IRQ10 | SCK6_A/DE6/USB_EXICEN/ET1_RXD1/ RGMII1_RXD1/RMII1_RXD1/CAT1_ERXD1 | GTIOC13A/ULPTEE1 | ADSYNC |
| A17 | — | P207 | — | IRQ25 | ET1_RXD5 | GTCPPO3 | — |
| A18 | VSS | — | — | — | — | — | — |
| B1 | — | P813 | SDCS | IRQ15 | SCK7_A/DE7 | GTETRGA/GTIOC7B | — |
| B2 | — | PA12 | D9/DQ9 | IRQ11 | RXD9_C/SCL9_C/MISO9_C | GTIW/GTIOC6B | — |
| B3 | — | P115 | D6/DQ6 | IRQ31-DS | CTS0_A/MOSIA_B/SD0DAT7_B | GTETRGD/GTIOC5A | AD0FLAG1 |
| B4 | — | PA11 | D8/DQ8 | IRQ10 | SCK9_C/DE9 | GTIV/GTIOC6A | — |
| B5 | — | P112 | D3/DQ3 | IRQ27 | TXD0_A/SDA0_A/MOSI0_A/SSLA2_B/ SD0DAT4_B | GTETRGA/GTIOC3B/ULPTOB0-DS | ADST0 |
| B6 | TDO/SWO/CLKOUT | P209 | — | IRQ25 | TXD9_B/SDA9_B/MOSI9_B/CTX1 | GTOVUP/GTIOC1A | — |
| B7 | TCK/SWCLK | P211 | — | IRQ23 | SCK9_B/DE9 | GTOUUP/GTIOC0A | — |
| B8 | VLO | — | — | — | — | — | — |
| B9 | VLO | — | — | — | — | — | — |
| B10 | VSS_DCDC | — | — | — | — | — | — |
| B11 | VCC_DCDC | — | — | — | — | — | — |
| B12 | VCC_DCDC | — | — | — | — | — | — |
| B13 | — | P311 | — | IRQ23-DS | SCK3_B/DE3/CRX0/ET1_TX_CLK/ CAT1_TX_CLK | GTADSM1/GTCPPO6/AGTOB1 | — |
| B14 | — | P908 | — | IRQ11 | TXD6_A/SDA6_A/MOSI6_A/CRX1/ USB_OVRCURB/ET1_RXD2/RGMII1_RXD2/ CAT1_ERXD2 | GTIOC12B/ULPTEVI1 | ADST1 |
| B15 | — | P909 | — | IRQ21-DS | RXD6_A/SCL6_A/MISO6_A/CTX1/ USB_OVRCURA/ET1_RXD3/RGMII1_RXD3/ CAT1_ERXD3 | GTIOC12A/ULPTOA1 | ADST0 |
| B16 | — | P904 | — | IRQ2 | ET1_RXD4 | GTIOC11B | — |
| B17 | — | PD01 | — | IRQ22 | SCK8_C/DE8/SD0DAT2_C/ET1_RXD6 | GTCPPO2 | — |
| B18 | — | PD02 | — | IRQ21 | TXD8_C/SDA8_C/MOSI8_C/SD0DAT1_C/ ET1_RXD7 | GTCPPO1 | — |
| C1 | — | PA06 | CS1/CKE | IRQ17 | CTS2_C/SD0DAT1_A | GTETRGC/GTIOC7B | — |
| C2 | — | P613 | D15/DQ15 | IRQ19 | CTS0_C | GTETRGA/GTIOC9B/AGTO1 | — |
| C3 | — | PA13 | D10/DQ10 | IRQ12 | CTS_RTS9_C/SS9_C/DE9 | GTOVUP/GTIOC10A | — |
| C4 | — | P300 | D2/DQ2 | IRQ4 | SCK0_A/DE0/SSLA3_B/SD0DAT3_B | GTIOC3A/ULPTEVI0-DS | — |
| C5 | — | P302 | D0/DQ0 | IRQ5 | RXD6_B/SCL6_B/MISO6_B/SD0DAT1_B | GTOUUP/GTIOC4A/ULPTO0-DS | — |
| C6 | — | P200 | — | NMI | — | — | — |
| C7 | RES | — | — | — | — | — | — |
| C8 | — | P110 | — | IRQ20 | SD0DAT4_C | GTIOC9B | — |
| C9 | — | P903 | — | IRQ1 | — | GTIOC11A | — |
| C10 | TCLK | P308 | — | IRQ26-DS | CTS3_B/SD0CLK_B/ET1_TX_ER/ ETHPHYCLK | GTIU/GTCPPO9/ULPTOB1 | — |
| C11 | TDATA2 | P305 | — | IRQ8 | SD0WP/ET1_TXD2/RGMII1_TXD2/ CAT1_ETXD2 | GTOVUP/GTCPPO12/ULPTEE1 | — |
| C12 | TDATA0 | P307 | — | IRQ27-DS | CTS_RTS6_A/SS6_A/DE6/SD0CMD_B/ ET1_TXD0/RGMII1_TXD0/RMII1_TXD0/ CAT1_ETXD0 | GTIV/GTCPPO10/ULPTOA1 | — |
| C13 | — | P911 | — | IRQ6 | ET1_TXD5 | GTIOC3B | — |

### Table 1.16 Pin list for the SiP product (3 of 9)

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| C14 | CLKOUT | P206 | CS7 | IRQ0-DS | USB_VBUSEN/SD0DAT7_C/ET1_RX_DV/ RGMII1_RX_CTL/RMII1_CRS_DV/ CAT1_RX_DV | GTIU/GTCPPO0/ULPTOB1 | — |
| C15 | — | PD04 | — | IRQ20 | CTS_RTS8_C/SS8_C/DE8/SD0CMD_C/ ET0_RXD5 | GTIOC3A | — |
| C16 | — | PD03 | — | IRQ21 | RXD8_C/SCL8_C/MISO8_C/SD0DAT0_C/ ET0_RXD4 | GTIOC3B | — |
| C17 | — | PD05 | — | IRQ19 | CTS8_C/SD0CLK_C/ET0_RXD6 | GTIOC2B | — |
| C18 | — | PD06 | — | IRQ18 | SD0WP/ET0_RXD7 | GTIOC2A | — |
| D1 | — | PA04 | A1 | IRQ19 | SCK2_C/DE2/SD0DAT3_A | GTIU/GTIOC4B | ADST0 |
| D2 | CACREF/CLKOUT | P611 | D13/DQ13 | IRQ17 | SCK0_C/DE0/MOSIA_B | GTOULO/GTIOC4B | — |
| D3 | — | P610 | D12/DQ12 | IRQ16 | RXD0_C/SCL0_C/MISO0_C/RSPCKA_B/ CRX1 | GTOUUP/GTIOC4A/ULPTOB1-DS | — |
| D4 | — | PA14 | D11/DQ11 | IRQ13 | TXD9_C/SDA9_C/MOSI9_C | GTOVLO/GTIOC10B | — |
| D5 | — | P303 | — | IRQ29-DS | SCK6_B/DE6 | GTIOC7B | — |
| D6 | — | P915 | — | IRQ8 | CTS6_B | GTIOC5A | — |
| D7 | — | P108 | — | IRQ24 | SD0DAT6_C | GTIOC10B | — |
| D8 | — | P111 | — | IRQ19 | SD0DAT3_C | GTIOC9A | — |
| D9 | — | P109 | — | IRQ23 | SD0DAT5_C | GTIOC10A | — |
| D10 | — | P310 | — | IRQ24-DS | TXD3_B/SDA3_B/MOSI3_B/ET1_TX_EN/ RGMII1_TX_CTL/RMII1_TX_EN/ CAT1_TX_EN | GTCPPO7/AGTEE1 | — |
| D11 | TDATA3 | P304 | — | IRQ9 | SD0DAT0_B/ET1_TXD3/RGMII1_TXD3/ CAT1_ETXD3 | GTOVLO/GTIOC7A/ULPTO1 | — |
| D12 | TDATA1 | P306 | — | IRQ28-DS | SD0CD/ET1_TXD1/RGMII1_TXD1/ RMII1_TXD1/CAT1_ETXD1 | GTIW/GTCPPO11/ULPTEVI1 | — |
| D13 | — | P912 | — | IRQ5 | ET1_TXD6 | GTIOC3A | — |
| D14 | — | PB04 | — | IRQ9 | SCK5_C/DE5/ET0_TXD3/RGMII0_TXD3/ CAT0_ETXD3 | GTCPPO3 | AD0FLAG1 |
| D15 | — | PB07 | — | IRQ1 | ET0_TXD5 | GTIOC9B | — |
| D16 | — | PB05 | — | IRQ15 | CTS5_C/ET0_TXD7 | GTCPPO4 | — |
| D17 | — | PB03 | — | IRQ13 | TXD5_C/SDA5_C/MOSI5_C/ET0_TXD2/ RGMII0_TXD2/CAT0_ETXD2 | GTCPPO1 | ADSYNC |
| D18 | — | PB01 | ALE | IRQ12 | CTS_RTS1_B/SS1_B/DE1/ET0_TX_CLK/ CAT0_TX_CLK | GTCPPO2 | AD1FLAG1 |
| E1 | — | PA15 | EBCLK/ SDCLK | IRQ14 | CTS9_C | GTIOC7A | — |
| E2 | — | P615 | — | IRQ7 | TXD7_A/SDA7_A/MOSI7_A | GTETRGC/GTCPPO10 | — |
| E3 | — | P614 | WR/WR0/ DQM0 | IRQ20 | RXD7_A/SCL7_A/MISO7_A | GTETRGB/GTCPPO9/AGTO0 | — |
| E4 | — | P612 | D14/DQ14 | IRQ18 | CTS_RTS0_C/SS0_C/DE0/SSLA0_B | GTIOC9A | — |
| E5 | — | P914 | — | IRQ9 | CTS_RTS6_B/SS6_B/DE6 | GTIOC5B | — |
| E6 | MD | P201 | — | IRQ4 | — | — | — |
| E10 | — | P902 | ALE | IRQ0 | ETHPHYCLK | GTCPPO13 | — |
| E11 | — | P312 | — | IRQ22-DS | CTS_RTS3_B/SS3_B/DE3/CTX0/ ET1_RX_ER/RMII1_RX_ER/CAT1_RX_ER | GTADSM0/GTCPPO5/AGTOA1 | — |
| E12 | — | P910 | — | IRQ7 | ET1_TXD4 | GTCPPO12 | — |
| E13 | CLKOUT | P913 | — | IRQ3 | ET1_TXD7 | GTCPPO11 | — |
| E14 | — | PB02 | — | IRQ11 | RXD5_C/SCL5_C/MISO5_C/ET0_TXD1/ RGMII0_TXD1/RMII0_TXD1/CAT0_ETXD1 | GTCPPO0 | ADST1 |
| E15 | — | PB06 | — | IRQ0 | CTS_RTS5_C/SS5_C/DE5/ET0_TXD6 | GTIOC9A | — |
| E16 | — | PD07 | — | IRQ17 | SD0CD/ET0_TXD4 | GTCPPO0 | — |
| E17 | — | PB00 | — | IRQ10 | SCK1_B/DE1/ET0_TXD0/RGMII0_TXD0/ RMII0_TXD0/CAT0_ETXD0/DSM1CLK2 | GTCPPO4 | ADST0 |
| E18 | — | P706 | — | IRQ7 | RXD1_B/SCL1_B/MISO1_B/ET0_GTX_CLK/ RGMII0_TXC/ETHPHYCLK/DSM1CLK0 | GTCPPO2/AGTIO0 | — |
| F1 | — | PA02 | A3 | IRQ31 | RXD2_C/SCL2_C/MISO2_C/SD0DAT5_A | GTIW/GTCPPO9 | ADSYNC |
| F2 | — | PA10 | CS2/RAS | IRQ4 | SCK5_B/DE5 | GTCPPO13 | — |
| F3 | — | PA08 | CS0/WE | IRQ6 | RXD5_B/SCL5_B/MISO5_B | GTETRGD/GTCPPO11 | — |

### Table 1.16 Pin list for the SiP product (4 of 9)

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| F4 | — | PA09 | CS3/CAS | IRQ5 | TXD5_B/SDA5_B/MOSI5_B | GTCPPO12 | — |
| F5 | — | PC14 | — | IRQ0 | TXD6_C/SDA6_C/MOSI6_C/ET0_WOL/ DSM1DAT1 | GTADSM1/GTCPPO9 | — |
| F7 | VCC_08 | — | — | — | — | — | — |
| F8 | VSS_08 | — | — | — | — | — | — |
| F9 | VSS3 | — | — | — | — | — | — |
| F10 | VCL3 | — | — | — | — | — | — |
| F11 | VSS_07 | — | — | — | — | — | — |
| F12 | VCC_07 | — | — | — | — | — | — |
| F13 | — | P700 | — | IRQ16-DS | RXD2_B/SCL2_B/MISO2_B/MISOA_C/ SD1WP/ET0_RXD2/RGMII0_RXD2/ CAT0_ERXD2/DSM0CLK0 | GTIOC5A | — |
| F14 | — | P702 | — | IRQ18-DS | CTS2_B/RSPCKA_C/SD1DAT5_B/ ET0_RXD0/RGMII0_RXD0/RMII0_RXD0/ CAT0_ERXD0/DSM0CLK2 | GTIOC6A/ULPTO0 | — |
| F15 | — | P406 | — | IRQ31 | TXD2_B/SDA2_B/MOSI2_B/SSLA3_C/ SD1CD/ET0_RXD3/RGMII0_RXD3/ CAT0_ERXD3/DSM0DAT2 | GTIOC1B | — |
| F16 | — | P701 | — | IRQ17-DS | CTS_RTS2_B/SS2_B/DE2/MOSIA_C/ SD1DAT4_B/ET0_RXD1/RGMII0_RXD1/ RMII0_RXD1/CAT0_ERXD1/DSM0CLK1 | GTIOC5B/ULPTO1 | — |
| F17 | — | P707 | — | IRQ8 | TXD1_B/SDA1_B/MOSI1_B/ET0_TX_ER/ ETHPHYCLK/DSM1CLK1 | GTCPPO3 | — |
| F18 | — | P705 | — | IRQ19 | CTS1_B/SSLA2_C/CRX0/ET0_TX_EN/ RGMII0_TX_CTL/RMII0_TX_EN/ CAT0_TX_EN/DSM1DAT2 | GTADSM1/GTCPPO1/AGTIO0 | — |
| G1 | — | PA00 | A5 | IRQ22 | CTS_RTS5_B/SS5_B/DE5/SD0DAT7_A | GTOVLO/GTCPPO7 | AD1FLAG1 |
| G2 | — | PA03 | A2 | IRQ20 | TXD2_C/SDA2_C/MOSI2_C/SD0DAT4_A | GTIV/GTCPPO10 | ADST1 |
| G3 | — | PA05 | A0/BC0/ DQM1 | IRQ18 | CTS_RTS2_C/SS2_C/DE2/SD0DAT2_A | GTETRGD/GTIOC4A | — |
| G4 | — | PA07 | RD | IRQ16 | CTS7_A/SD0DAT0_A | GTETRGB/GTIOC7A | VCOUT |
| G5 | — | PC12 | — | IRQ2 | SCK6_C/DE6/ET0_MDIO/CAT0_MDIO/ DSM1CLK0 | GTCPPO11 | — |
| G7 | VCC_09 | — | — | — | — | — | — |
| G8 | VSS_09 | — | — | — | — | — | — |
| G9 | VSS4 | — | — | — | — | — | — |
| G10 | VCL4 | — | — | — | — | — | — |
| G11 | VSS_06 | — | — | — | — | — | — |
| G12 | VCC_06 | — | — | — | — | — | — |
| G13 | — | P405 | — | IRQ30 | SCK2_B/DE2/SD1DAT3_B/ET0_RX_DV/ RGMII0_RX_CTL/RMII0_CRS_DV/ CAT0_RX_DV/DSM0DAT1 | GTIOC1A/AGTIO1 | — |
| G14 | — | P704 | — | IRQ26 | SSLA1_C/CTX0/SD1DAT7_B/ET0_RX_ER/ RMII0_RX_ER/CAT0_RX_ER/DSM1DAT1 | GTADSM0/GTCPPO0/AGTO0 | — |
| G15 | — | P703 | — | IRQ19-DS | SSLA0_C/SD1DAT6_B/ET0_RX_CLK/ RGMII0_RXC/RMII0_REF50CK/ CAT0_RX_CLK/DSM1DAT0 | GTIOC6B/AGTO1 | VCOUT |
| G16 | — | PB14 | — | IRQ10 | CATSYNC0 | — | — |
| G17 | — | PB12 | — | IRQ6 | CATLINKACT0 | — | — |
| G18 | — | PB11 | — | IRQ5 | CATLEDERR | — | — |
| H1 | — | P504 | A7 | IRQ7 | SD0WP | GTOULO/GTCPPO1 | — |
| H2 | — | P503 | A6 | IRQ6 | SD0CD | GTOUUP/GTCPPO6 | — |
| H3 | — | P505 | A8 | IRQ8 | SD0CLK_A | GTOWUP/GTCPPO2 | — |
| H4 | — | PA01 | A4 | IRQ21 | CTS5_B/SD0DAT6_A | GTOVUP/GTCPPO8 | AD0FLAG1 |
| H5 | — | PC11 | — | IRQ3 | CTS_RTS6_C/SS6_C/DE6/ET0_MDC/ CAT0_MDC/DSM1CLK1 | GTCPPO12 | — |
| H7 | VCC_10 | — | — | — | — | — | — |
| H8 | VSS_10 | — | — | — | — | — | — |
| H9 | VSS7 | — | — | — | — | — | — |
| H10 | VCL5 | — | — | — | — | — | — |

### Table 1.16 Pin list for the SiP product (5 of 9)

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| H11 | VSS5 | — | — | — | — | — | — |
| H12 | VCC_04 | — | — | — | — | — | — |
| H13 | VSS_04 | — | — | — | — | — | — |
| H15 | VCC_05 | — | — | — | — | — | — |
| H16 | — | PB09 | — | IRQ13 | CATIRQ | — | — |
| H17 | — | PB10 | — | IRQ4 | CATLEDSTER | — | — |
| H18 | — | PB13 | — | IRQ7 | CATLINKACT1 | — | — |
| J1 | — | P506 | A9 | IRQ9 | SD0CMD_A | GTOWLO/GTCPPO3 | — |
| J2 | — | P507 | A10 | IRQ10 | CTS_RTS7_A/SS7_A/DE7/ET_TAS_STA0 | GTADSM0/GTIOC0A | — |
| J3 | — | P508 | A11 | IRQ1 | CTS5_A/ET_TAS_STA1 | GTADSM1/GTIOC0B | — |
| J4 | — | P509 | A12 | IRQ2 | CTS_RTS5_A/SS5_A/DE5/ET_TAS_STA2 | GTIOC1A/ULPTEVI1 | — |
| J5 | — | PC13 | — | IRQ1 | RXD6_C/SCL6_C/MISO6_C/ET0_INT/ DSM1DAT2 | GTCPPO10 | — |
| J7 | VCC2_11 | — | — | — | — | — | — |
| J8 | VSS_11 | — | — | — | — | — | — |
| J9 | VCL7 | — | — | — | — | — | — |
| J10 | VCL6 | — | — | — | — | — | — |
| J11 | VSS6 | — | — | — | — | — | — |
| J12 | VCL2 | — | — | — | — | — | — |
| J13 | VSS2 | — | — | — | — | — | — |
| J15 | — | PB08 | — | IRQ12 | CATLEDRUN | — | — |
| J16 | VSS_05 | — | — | — | — | — | — |
| J17 | VSS_03 | — | — | — | — | — | — |
| J18 | VSS_02 | — | — | — | — | — | — |
| K1 | — | PC15 | A16 | IRQ30 | CTS6_C/CRX1 | GTADSM0 | — |
| K2 | CACREF | P608 | A14 | IRQ22 | TXD5_A/SDA5_A/MOSI5_A | GTOWUP/GTCPPO4 | — |
| K3 | — | P510 | A13 | IRQ3 | RXD5_A/SCL5_A/MISO5_A/ET_TAS_STA3 | GTIOC1B/ULPTEVI0 | — |
| K4 | — | PD00 | A15 | IRQ23 | SCK5_A/DE5/CTX1 | GTOWLO/GTCPPO5 | — |
| K5 | VSS | — | — | — | — | — | — |
| K6 | VSS | — | — | — | — | — | — |
| K7 | VSS_12 | — | — | — | — | — | — |
| K8 | VSS9 | — | — | — | — | — | — |
| K9 | VCL9 | — | — | — | — | — | — |
| K10 | VCL8 | — | — | — | — | — | — |
| K11 | VSS8 | — | — | — | — | — | — |
| K12 | VCL1 | — | — | — | — | — | — |
| K13 | VSS1 | — | — | — | — | — | — |
| K15 | VCC_03 | — | — | — | — | — | — |
| K16 | VCC_02 | — | — | — | — | — | — |
| K17 | XTAL | P213 | — | IRQ2 | TXD1_C/SDA1_C/MOSI1_C | GTETRGC/GTIOC0A/ULPTEE0 | ADTRG1 |
| K18 | EXTAL | P212 | — | IRQ3 | RXD1_C/SCL1_C/MISO1_C | GTETRGD/GTIOC0B/AGTEE1 | — |
| L1 | — | PC10 | — | IRQ4 | DSM1CLK2 | GTCPPO13 | — |
| L2 | VSS | — | — | — | — | — | — |
| L3 | PUP | — | — | — | — | — | — |
| L4 | VCC2_16 | — | — | — | — | — | — |
| L5 | VSS_16 | — | — | — | — | — | — |
| L7 | VCC2_12 | — | — | — | — | — | — |
| L8 | VSS_14 | — | — | — | — | — | — |
| L9 | VSS_15 | — | — | — | — | — | — |
| L10 | VSS10 | — | — | — | — | — | — |
| L11 | VCL10 | — | — | — | — | — | — |

### Table 1.16 Pin list for the SiP product (6 of 9)

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| L12 | VCL0 | — | — | — | — | — | — |
| L13 | VSS0 | — | — | — | — | — | — |
| L14 | — | P403 | — | IRQ14-DS | CTS_RTS1_A/SS1_A/DE1/SD1DAT1_B/ ET1_WOL/CATI2CCLK | GTIOC3A/RTCIC1 | AD0FLAG1 |
| L15 | — | P404 | — | IRQ15-DS | CTS1_A/SD1DAT2_B/ET0_WOL/ CATI2CDATA/DSM0DAT0 | GTIOC3B/RTCIC2 | AD1FLAG1 |
| L16 | VCC_01 | — | — | — | — | — | — |
| L17 | XCOUT | P214 | — | IRQ21 | — | — | — |
| L18 | XCIN/EXCIN | P215 | — | IRQ20 | — | — | — |
| M1 | — | PC09 | — | IRQ5 | — | — | — |
| M2 | VSS | — | — | — | — | — | — |
| M3 | VSS | — | — | — | — | — | — |
| M4 | VCC2_17 | — | — | — | — | — | — |
| M5 | VSS_17 | — | — | — | — | — | — |
| M8 | VCC2_14 | — | — | — | — | — | — |
| M9 | VCC2_15 | — | — | — | — | — | — |
| M11 | VSS11 | — | — | — | — | — | — |
| M12 | VCL11 | — | — | — | — | — | — |
| M14 | — | P414 | A23 | IRQ9 | RXD4_B/SCL4_B/MISO4_B/SSLB0_B/CRX1/ ET1_MDIO/CATLATCH1 | GTIOC0B | — |
| M15 | CACREF | P402 | — | IRQ4-DS | SCK1_A/DE1/CRX0/SD1DAT0_B/ ET0_LINKSTA/CAT0_LINKSTA | RTCIC0 | — |
| M16 | — | P410 | A19 | IRQ5 | SCK3_A/DE3/SCL0_A[*1]/USB_OVRCURB-DS/GPTP_MATCH0/CATI2CCLK | GTOVLO/GTIOC9B/AGTOB1 | ADST0 |
| M17 | VBATT | — | — | — | — | — | — |
| M18 | VSS_01 | — | — | — | — | — | — |
| N1 | — | PC08 | — | IRQ29 | DSM1DAT0 | GTCPPO8 | — |
| N2 | VSS | — | — | — | — | — | — |
| N3 | VSS | — | — | — | — | — | — |
| N4 | VCC2_18 | — | — | — | — | — | — |
| N5 | VSS_18 | — | — | — | — | — | — |
| N7 | — | P105 | — | IRQ0 | CTS_RTS8_B/SS8_B/DE8/SSLB2_A/ OM_0_ECSINT1/DSM0DAT0 | GTIOC1A/ULPTO1-DS | ADSYNC |
| N10 | — | P810 | — | IRQ21 | SCK7_B/DE7/SD1DAT2_A | GTIOC10A/ULPTOA0 | — |
| N14 | — | P710 | CS5 | IRQ17 | CTS4_B/SSLB3_B/ET0_LINKSTA/ CAT0_LINKSTA | GTIOC11B | — |
| N15 | CACREF | P411 | A20 | IRQ4 | CTS_RTS3_A/SS3_A/DE3/USB_ID/ GPTP_PTPOUT1 | GTOVUP/GTIOC9A/AGTOA1 | — |
| N16 | — | P408 | A17 | IRQ7 | RXD3_A/SCL3_A/MISO3_A/SCL0_B[*1]/ USB_VBUSEN/GPTP_PTPOUT2/ CATRESETOUT | GTOWLO/GTIOC10A/ULPTOB0 | ADSYNC |
| N17 | — | P412 | A21 | IRQ20-DS | CTS3_A/USB_EXICEN/GPTP_PTPOUT0 | GTOULO/GTCPPO8/AGTEE1 | — |
| N18 | — | P401 | — | IRQ5-DS | RXD1_A/SCL1_A/MISO1_A/I3C_SDA0/ CTX0/SD1CMD_B | GTETRGA/GTIOC6B | — |
| P1 | VSS | — | — | — | — | — | — |
| P2 | VSS | — | — | — | — | — | — |
| P3 | VSS | — | — | — | — | — | — |
| P4 | VCC2_19 | — | — | — | — | — | — |
| P5 | VSS_19 | — | — | — | — | — | — |
| P6 | — | P104 | — | IRQ1 | CTS9_A/SSLB1_A/OM_0_CS1/DSM0DAT1 | GTETRGB/GTIOC1B | AD0FLAG1 |
| P7 | — | P107 | — | IRQ31 | CTS4_A/OM_0_CS0 | GTOWUP/GTIOC8A/AGTOA0 | ADST0 |
| P8 | — | P106 | — | IRQ16 | CTS8_B/SSLB3_A/OM_0_RESET | GTOWLO/GTIOC8B/AGTOB0/ ULPTEE1-DS | ADST1 |
| P9 | — | P811 | — | IRQ22 | CTS7_B/USB_ID/SD1DAT3_A | GTIOC10B/ULPTOB0 | — |
| P10 | — | P013 | — | IRQ14 | — | — | AN013 |
| P11 | — | P011 | — | IRQ16 | — | — | AN011 |

## Table 1.16 Pin list for the SiP product (7 of 9)

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| P12 | — | P807 | — | IRQ11 | — | GTIOC13A | — |
| P13 | CACREF | P708 | WR1/BC1 | IRQ11 | SCK4_B/DE4/SDA2_A[*1]/MOSIB_B/ ET0_MDC/CAT0_MDC | GTCPPO6 | — |
| P14 | — | P712 | — | IRQ2 | CTS1_C/SSLB1_B/GPTP_CAPTURE1/ CATLATCH1 | GTIOC2B/AGTOB0 | — |
| P15 | — | P714 | — | IRQ13 | TXD4_C/SDA4_C/MOSI4_C/GPTP_PPS1/ CATSYNC1 | GTIOC12B | — |
| P16 | — | P711 | — | IRQ3 | CTS_RTS1_C/SS1_C/DE1/SSLB2_B/ GPTP_PPS0/CATRESETOUT | GTIOC11A/AGTEE0 | — |
| P17 | — | P713 | — | IRQ14 | CTS4_C/GPTP_MATCH1/CATLATCH0 | GTIOC2A/AGTOA0 | — |
| P18 | — | P400 | — | IRQ0 | TXD1_A/SDA1_A/MOSI1_A/I3C_SCL0/ SD1CLK_B | GTIOC6A/AGTIO1 | ADTRG1 |
| R1 | — | P602 | — | IRQ28 | RXD0_B/SCL0_B/MISO0_B | GTIOC7B/ULPTEE0 | — |
| R2 | VSS | — | — | — | — | — | — |
| R3 | VSS | — | — | — | — | — | — |
| R4 | CACREF | P600 | — | IRQ30 | OM_0_RSTO1 | GTIOC6B/ULPTEVI1-DS | — |
| R5 | — | P601 | — | IRQ29 | SCK0_B/DE0/OM_0_WP1 | GTIOC6A/ULPTEVI0/RTCOUT | — |
| R6 | — | P102 | — | IRQ17 | TXD9_A/SDA9_A/MOSI9_A/RSPCKB_A/ CRX0/OM_0_SIO4/DSM0CLK0 | GTOWLO/GTIOC2B/AGTO0 | ADTRG0 |
| R7 | — | P801 | — | IRQ12 | TXD2_A/SDA2_A/MOSI2_A/OM_0_DQS/ DSM1DAT1 | GTIV/GTIOC11B/AGTOB0 | — |
| R8 | — | P803 | — | IRQ19 | SCK2_A/DE2/OM_0_SIO1/DSM1CLK0 | GTETRGC/GTIOC12B | — |
| R9 | — | P812 | — | IRQ23 | CTS_RTS7_B/SS7_B/DE7/USB_EXICEN/ SD1DAT4_A | GTIOC11A | AN022 |
| R10 | — | P012 | — | IRQ15 | — | — | AN012 |
| R11 | — | P010 | — | IRQ14 | — | — | AN010 |
| R12 | — | P009 | — | IRQ13-DS | — | — | AN009/ IVREF1 |
| R13 | — | P805 | — | IRQ30 | TXD8_A/SDA8_A/MOSI8_A/ET1_MDIO | — | AN017/ IVCMP0 |
| R14 | — | P512 | — | IRQ14 | CTS8_A/SCL1_A[*1]/CTX1/ET1_INT | GTIOC0A | — |
| R15 | — | P413 | A22 | IRQ18 | ET_TAS_STA3 | GTOUUP/GTCPPO7/ULPTEE1 | — |
| R16 | — | P515 | — | IRQ12 | CTS_RTS4_C/SS4_C/DE4/SCL2_B[*1]/ ET_TAS_STA0 | GTIOC13A | — |
| R17 | — | P709 | CS4 | IRQ10 | CTS_RTS4_B/SS4_B/DE4/SCL2_A[*1]/ MISOB_B/ET0_MDIO/CAT0_MDIO | GTCPPO5 | — |
| R18 | — | P407 | CS6 | IRQ22 | SCK1_C/DE1/SDA0_B[*1]/USB_VBUS/ GPTP_PTPOUT3/CATSYNC1 | GTIOC10B/AGTIO0/RTCOUT | ADTRG0 |
| T1 | Vpp | — | — | — | — | — | — |
| T2 | VSS | — | — | — | — | — | — |
| T3 | — | P315 | — | IRQ29 | SCK3_C/DE3/SSLA3_A | — | — |
| T4 | — | P900 | — | IRQ30 | CTS3_C | GTADSM0 | — |
| T5 | — | P103 | — | IRQ16 | CTS_RTS9_A/SS9_A/DE9/SSLB0_A/CTX0/ OM_0_SIO2/DSM0DAT2 | GTOWUP/GTIOC2A | AD1FLAG1 |
| T6 | — | P101 | — | IRQ1 | RXD9_A/SCL9_A/MISO9_A/MOSIB_A/ OM_0_SIO3/DSM0CLK1 | GTETRGB/GTIOC8A/AGTEE0 | — |
| T7 | — | P802 | — | IRQ18 | RXD2_A/SCL2_A/MISO2_A/OM_0_SIO6/ DSM1DAT2 | GTIW/GTIOC12A | — |
| T8 | — | P804 | — | IRQ14 | CTS_RTS2_A/SS2_A/DE2/OM_0_SIO7/ DSM1CLK1 | GTETRGD/GTIOC13A | — |
| T9 | — | P501 | — | IRQ25 | TXD8_B/SDA8_B/MOSI8_B/ USB_OVRCURA/SD1DAT6_A | GTIOC12A | AN020 |
| T10 | AVCC0 | — | — | — | — | — | — |
| T11 | AVSS0 | — | — | — | — | — | — |
| T12 | — | P005 | — | IRQ10-DS | — | — | AN005/ IVCMP3 |
| T13 | — | P003 | — | IRQ29 | — | — | AN003/ IVCMP3 |
| T14 | — | P513 | — | IRQ31 | SCK8_A/DE8/ET0_INT | GTIOC13B | AN016/ IVCMP0 |

**Table 1.16 Pin list for the SiP product (8 of 9)**

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| T15 | — | P514 | — | IRQ13 | SCK4_C/DE4/SDA2_B*1/ET_TAS_STA1 | GTIOC13B | — |
| T16 | — | P415 | WAIT | IRQ8 | TXD4_B/SDA4_B/MOSI4_B/RSPCKB_B/ CTX1/ET1_MDC/CATLATCH0 | GTIOC0A | — |
| T17 | — | P409 | A18 | IRQ6 | TXD3_A/SDA3_A/MOSI3_A/SDA0_A*1/ USB_OVRCURA-DS/GPTP_CAPTURE0/ CATI2CDATA | GTOWUP/ULPTOA0 | ADST1 |
| T18 | VCC_USB | — | — | — | — | — | — |
| U1 | VCC2_13 | — | — | — | — | — | — |
| U2 | CLKOUT | P205 | — | IRQ1-DS | TXD4_A/SDA4_A/MOSI4_A/SCL1_B*1/ SSLA1_A/USB_OVRCURA/SD1CD | GTIV/GTIOC4A/AGTO1 | — |
| U3 | — | P203 | — | IRQ2-DS | RXD4_A/SCL4_A/MISO4_A/RSPCKA_A/ CTX0/USB_VBUSEN/SD1CLK_A | GTIOC5A/ULPTOA1 | — |
| U4 | — | P313 | — | IRQ27 | TXD3_C/SDA3_C/MOSI3_C/MISOA_A/ USB_ID/SD1DAT0_A | — | — |
| U5 | — | P901 | — | IRQ31 | CTS_RTS3_C/SS3_C/DE3 | GTADSM1/AGTIO1 | — |
| U6 | — | P809 | — | IRQ20 | TXD7_B/SDA7_B/MOSI7_B/OM_0_SCLKN | — | — |
| U7 | — | P800 | — | IRQ11 | CTS2_A/OM_0_SIO5/DSM1DAT0 | GTIU/GTIOC11A/AGTOA0 | — |
| U8 | — | P502 | — | IRQ26 | SCK8_B/DE8/USB_OVRCURB/SD1DAT7_A | GTIOC12B | AN019 |
| U9 | — | P014 | — | IRQ27 | — | — | AN014/DA0/ IVCMP0 |
| U10 | VREFL | — | — | — | — | — | — |
| U11 | VREFL0 | — | — | — | — | — | — |
| U12 | — | P004 | — | IRQ9-DS | — | — | AN004/ IVCMP2 |
| U13 | — | P007 | — | IRQ28 | — | — | AN007/ IVCMP3 |
| U14 | — | P001 | — | IRQ7-DS | — | — | AN001/ IVCMP3 |
| U15 | — | P806 | — | IRQ0 | RXD8_A/SCL8_A/MISO8_A/ET1_MDC | — | AN018 |
| U16 | — | P715 | — | IRQ12 | RXD4_C/SCL4_C/MISO4_C/ET_TAS_STA2 | GTIOC12A | — |
| U17 | — | P815 | — | IRQ15 | CTX0/USB_DM | GTIOC8A | — |
| U18 | VSS_USB | — | — | — | — | — | — |
| V1 | VSS | — | — | — | — | — | — |
| V2 | CACREF | P204 | — | IRQ26 | SCK4_A/DE4/SDA1_B*1/SSLA0_A/ USB_OVRCURB/SD1WP | GTIW/GTIOC4B/AGTIO1 | — |
| V3 | — | P202 | — | IRQ3-DS | CTS_RTS4_A/SS4_A/DE4/MOSIA_A/CRX0/ USB_EXICEN/SD1CMD_A | GTIOC5B/ULPTOB1 | — |
| V4 | — | P314 | — | IRQ28 | RXD3_C/SCL3_C/MISO3_C/SSLA2_A/ SD1DAT1_A | — | ADTRG0 |
| V5 | VSS_13 | — | — | — | — | — | — |
| V6 | — | P808 | — | IRQ15 | RXD7_B/SCL7_B/MISO7_B/OM_0_SCLK/ DSM1CLK2 | GTIOC13B | — |
| V7 | — | P100 | — | IRQ2 | SCK9_A/DE9/MISOB_A/OM_0_SIO0/ DSM0CLK2 | GTETRGA/GTIOC8B/AGTIO0 | — |
| V8 | CACREF | P500 | — | IRQ24 | RXD8_B/SCL8_B/MISO8_B/USB_VBUSEN/ SD1DAT5_A | GTIOC11B | AN021 |
| V9 | — | P015 | — | IRQ13 | — | — | AN015/DA1/ IVCMP0 |
| V10 | VREFH | — | — | — | — | — | — |
| V11 | VREFH0 | — | — | — | — | — | — |
| V12 | — | P008 | — | IRQ12-DS | — | — | AN008/ IVREF0 |
| V13 | — | P006 | — | IRQ11-DS | — | — | AN006/ IVCMP2 |
| V14 | — | P000 | — | IRQ6-DS | — | — | AN000/ IVCMP2 |
| V15 | — | P002 | — | IRQ8-DS | — | — | AN002/ IVCMP2 |
| V16 | — | P511 | — | IRQ15 | CTS_RTS8_A/SS8_A/DE8/SDA1_A*1/CRX1/ ET1_LINKSTA/CAT1_LINKSTA | GTIOC0B | — |
| V17 | — | P814 | — | IRQ16 | CRX0/USB_DP | GTIOC8B | — |

**Table 1.16 Pin list for the SiP product (9 of 9)**

| BGA303 | Power, System, Clock, Debug, CAC | I/O ports | ExBus/ SDRAM | Ex.Interrupt | SCI/IIC/I3C/SPI/CANFD/USBFS/OSPI/ SDHI/MMC/ESWM(GMII, RGMII, MII, RMII)/ EtherCAT/DSMIF | GPT/AGT/ULPT/RTC | ADC16H/ DAC12/ ACMPHS |
|---|---|---|---|---|---|---|---|
| V18 | VSS | — | — | — | — | — | — |

Note: Several pin names have the added suffix of _A, _B, and _C. These suffixes have special conditions for electrical characteristics. See section 60, Electrical Characteristics for detail.

Note 1. There are two types for IIC function, one is a simple IIC by SCI and another is a dedicated IIC. This terminal is for a dedicated IIC.